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# Thinking outside the container: A sparse partial least squares approach to forecasting trade flows ☆

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ABSTRACT

Global container ship movements may reliably predict trade flows. First, this paper provides the methodology to construct maritime shipping time series from a dataset comprising millions of container vessel positions annually. Second, to forecast monthly goods trade using these time series, this study outlines the use of the least absolute shrinkage and selection operator (LASSO) in combination with a partial least squares process (PLS). An expanding window, out-of-sample exercise demonstrates that constructed forecasts outperform benchmark models for the vast majority of 76 countries and regions. The performance holds true for unilateral and bilateral trade flows, for trade of developed and developing countries, for real and nominal trade, as well as for time periods of economic crisis such as the COVID-19 pandemic. The resulting forecasts of trade flows precede official statistics by several months and may facilitate quantification of supply chain disruptions and trade wars.

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## 1. Introduction

As the coronavirus spread across the globe, it disrupted the goods trade in its wake. In April 2020, for instance, Germany and France experienced year-over-year export drops of 33.6% and 44.5%, respectively (WTO, 2022). Even two years after the first outbreak of the novel virus, supply chain disruptions continued to limit the economic recovery. Celasun, Mineshima, Hansen, Zhou, and Spector (2022) find that the euro area's GDP in 2021 would have been 2% higher in the absence of supply shortages. To mitigate the uncertainty associated with plummeting trade volumes, decision-makers in governments and corporations require as recent information as possible. Yet, even the most advanced statistical offices publish unilateral trade data with a lag of several weeks and bilateral data often with a lag of months.

This study applies machine learning techniques to container ship movements. Container vessels moved more than 8 trillion USD worth of goods across the seas in 2020 (UNCTAD, 2021), accounting for almost half of global trade by value. Container ship positions are also available in a matter of hours. This study obtains ship information from January 2015 to August 2022 for the global fleet of container ships. The raw data has two distinct formats: The first set of data is an exhaustive set of all 450 thousand port calls per year, i.e., the stops of container ships at ports to load or unload cargo. In addition, this study processes over 1,7 Million daily container ship positions per year, which can be aggregated by geographic position and course using a clustering algorithm. Overall, the data yields several thousand time

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series of container ship activity. Variable selection with LASSO (least absolute shrinkage and selection operator) reduces the shipping data to a smaller set of predictor time series, and a factor model called partial least squares algorithm (PLS) estimates trade flows. This procedure generates accurate forecasts of trade flows one month ahead, which predate official statistics by several months. For several countries, forecasts may hold predictive power up to five months ahead. Expanding window, out-of-sample tests show that the forecasts significantly outperform and encompass benchmark models for the vast majority of imports and exports of 76 countries and regions, as well as bilateral trade flows between major economies. The combination of LASSO and PLS is at least as accurate in estimations as related machine learning tools.

The literature on trade flow prediction is relatively young but is growing rapidly. Veenstra and Haralambides (2001) provide an early work for predicting annual commodity flows using a vector autoregression (VAR) model. In a study on quarterly imports of OECD-25 countries, Keck, Raubold, and Truppia (2010) use an autoregressive integrated moving average model (ARIMA) and GDP growth to predict import growth. D'Agostino, Modugno, and Osbat (2017) develop factor models for monthly bilateral trade flows of the euro area utilizing macroeconomic data such as purchasing manager indices (PMIs) and consumer price indices (CPIs), among others. Similarly, Lehmann (2021) study the importance of survey-based indicators such as production expectations and industrial confidence indicators for forecasting quarterly exports of European countries. In a proof-of-concept study, Cerdeiro, Komaromi, Liu, and Saeed (2020) elaborate methods to estimate ships' cargo in metric tons and show that aggregated ship cargo tends to correlate with countries' imports and exports in the same period. Cerdeiro and Komaromi (2022) also employ this dataset to identify the supply spillovers from lockdowns causally. Furthermore, Circlaeys, Kanitkar, and Kumazawa (2017) employ neural networks to validate annual bilateral trade using GDP values, distances, and other control variables from the same period. Lastly, Dubovik et al. (2022), Kim (2020) use neural networks and deep learning techniques to forecast South Korean exports and total world trade, respectively. This study combines and advances the above studies as it processes ship positions to approximate raw cargo flows. It uses machine learning techniques to make forecasts for the current month and one month ahead of headline trade statistics. To the best of the author's knowledge, this is the first paper to introduce forecast methods for imports and exports of 76 countries and regions, as well as bilateral trade flows, and to compare them in out-of-sample exercises to benchmark models.

In terms of methodology, this paper follows a growing literature using the statistical procedures frequently referred to as "factor models" to reduce dimensionality (see Bai & Ng, 2004; Jurado, Ludvigson, & Ng, 2015; Stock & Watson, 2002). For instance, Eickmeier and Ng (2011) find that PLS is well suited to integrate rich international indicators to model national GDP. Similarly, Camacho and Perez-Quiros (2010) successfully forecast euro area GDP using a small-scale factor model that dramatically limits the number of predictors. As the number of predictors in this study is high even for factor models, explanatory time series are pre-selected using the LASSO as introduced by Tibshirani (1996) and a custom algorithm taking into account the magnitude and geographic proximity of container ship ports. The methodologically most closely related study is Fuentes, Poncela, and Rodríguez (2015). In their inflation forecast models, the authors refer to their predictor selection procedure as sparse partial least squares (SPLS). This paper contributes to this strand of econometric methodology by combining LASSO with the PLS statistical procedure and outlining aggregation methods for geocoded data. Additionally, the above studies combine explanatory time series from various sources to forecast one dependent time series. This work flips this approach: The aim is to provide forecasts for many trade flows from one source of information, the container shipping network. National practitioners can then integrate these forecasts into country-specific model combinations.

Lastly, this study supplements economic applications of the Automatic Identification System (AIS), which records ship positions and other information based on ships' radio signals. For other uses of AIS data in economics, see, for instance, Brancaccio, Kalouptsidi, and Papageorgiou (2017), Heiland, Moxnes, Ulltveit-Moe, and Zi (2019), Sandkamp, Stamer, and Yang (2022) and Ganapati, Wong, and Ziv (2020). Arslanalp, Marini, and Tumbarello (2019) approximate cargo flows from the activity of two Maltesian ports and compare it to Malta's official trade data. Cerdeiro et al. (2020) develop methods to calculate port calls from ship positions and show that aggregate cargo flows tend to correlate with countries' trade. This study does not calculate port calls but receives pre-calculated port calls from a maritime intelligence provider. Instead, this work overlays AIS-derived predictor time series with a machine learning algorithm for forecasting. These algorithms not only trace a country's seaborne actual cargo flows but also exploit any source of correlation arising through common trends of countries, input–output linkages, and substitution and complementary relationships that transcend borders.

The following section describes the partial least squares procedure and the required time series tests. Section 3 outlines converting AIS data to independent time series. The chapter also validates pre-calculated port calls and outlines the leading indicators used in the benchmark model. Results are presented in Section 4 and include benchmark comparisons and encompassing tests. The section separately tests forecasts for real unilateral trade flows and bilateral trade, and distinguishes between normal time periods and times of economic crisis. Furthermore, it presents a comparison to other estimation and variable selection techniques. Section 5 discusses the robustness of results to measurement error. Various robustness checks show that results hold for predicting nominal and real trade and limiting input data to the first half of the month. The last chapter concludes and gives an outlook for work ahead.

## 2. Methodology

### 2.1. Shrinkage and partial least squares

This subsection outlines the methods to reduce the dimensionality of the predictor space and calculate factors for estimations using the partial least squares procedure. Wold (1975) introduced the partial least squares (PLS) technique as early as 1975, and several handbooks describe the technique in detail (see, for instance, Esposito Vinzi et al., 2010; Garthwaite, 1994; Haenlein & Kaplan, 2004; Lohmöller, 2013). Statistical procedures such as the principle component regression (PCR) and PLS gain popularity because they reduce dimensionality in the span of the predictor time series in the presence of strong colinearity among predictors (Wold, Sjöström, & Eriksson, 2001). As the number of predictors (N) and the number of time series observations (T) tend towards infinity, PLS makes consistent estimates of the factor space under the following assumptions: Estimation errors are stationary, components have non-trivial loadings, and errors have at most weak cross-correlation (Bai & Ng, 2002; Stock & Watson, 2002). Chun and Keleş (2010) relax the assumption of $N$ and $T$ going to infinity. The authors establish that consistent estimates are achieved if and only if $N$ grows more slowly than T. Breitung and Hansen (2021) find in Monte Carlo simulations that $T = 10$ is sufficient to achieve reasonably unbiased estimates for PCR. However, Rönkkö (2014) caution that the likelihood of correlations by chance increases as $N$ grows.

This has several implications for this study: First, the out-of-sample study becomes indispensable since a correlation by chance between dependent and independent variables leads to a modeling error. Since the out-of-sample approach divides the data into a training and a testing set, the out-of-sample test will report the modeling errors stemming from erroneous correlation in the training phase. Second, the smallest length of the expanding window is chosen to be reasonably large to avoid bias. In practice, the expanding window analysis begins at half the time periods available so that $T$ is greater or equal to 46. Second, the number of predictor time series $N$ must be reduced below the number of observations $T$ before running PLS. This is partly achieved in the aggregation of data, as outlined in the section below. Additionally, the individual predictor time series, $x_{tj}$, for $j = 1, \ldots, N$, are pre-selected for the dependent trade flow series, $y_{t+h}$, at forecast horizon, h, using the least absolute shrinkage and selection operator (LASSO) introduced by Tibshirani (1996). The LASSO estimate for coefficients $\beta = (\beta_1, \ldots, \beta_N)^T$ is defined as

$$\hat{\beta} = \arg\min_{\beta} \left[ \sum_{t=1}^{T} \left( y_{t+h} - \sum_{j=1}^{N} \beta_j x_{tj} \right)^2 + \lambda \sum_{j=1}^{N} |\beta_j| \right] \quad \text{subject to } \lambda \geq 0 \tag{1}$$

where penalization for additional coefficients is calculated by the sum of the absolute coefficients multiplied by the shrinkage intensity λ. Beginning with a value of $\lambda = 1$, the parameter is iteratively adjusted so that, ultimately, 35 to 45 coefficients have a non-zero value. The predictor time series corresponding to non-zero coefficient values are selected. Overall, this reduces the number of predictor time series from $N \approx 640$ to $J \in \{35, \ldots, 45\}$, thereby below the minimum value of $T = 46$ in the expanding window analysis. The procedure is repeated for every dependent time series and forecast horizon. In the partial least squares procedure, predictor time series can also receive a zero weight. Hence, the partial least squares approach will not be affected even if the lower bound of 35 is still too high for a factor model that minimizes forecast errors. Table 9 shows pre-selected predictor time series for US-American import nowcasts. Among the 36 predictor time series selected, the list contains several sea areas along the North American coasts, the ports of Miami, New York, and Jackson, and an aggregate of all US ports. Also, several areas and ports in the North Sea and in proximity to the Panama Canal are selected.

The following summarizes the partial least squares process: Statistical procedures such as PLS rely on the idea that a small number of latent factors $f_{tk}$ capture variation for all explanatory time series $x_{tj}$. The number of latent factors $K$ can be much smaller than the number of predictors $J$. Algorithms such as PLS and PCR approximate these latent factors and use them as components in a model for the dependent time series $y_{t+h}$.

$$x_{tj} = \sum_{k=1}^{K} \delta_{jk} f_{tk} + \varepsilon_{tj}, \qquad \forall x_{tj} \tag{2}$$

$$y_{t+h} = \alpha + \sum_{k=1}^{K} \beta_k \widehat{f}_{tk} + \eta_{t+h} \tag{3}$$

where $x_{tj}$ is a predictor time series, $f_{tk}$ refers to latent variables (also known as "factors" or "components"), $h$ is the forecast horizon and $\varepsilon_{tj}$ and $\eta_{t+h}$ are unobserved error terms with finite variance and zero mean. The PLS algorithm estimates the so-called factor loadings $\delta_{jk}$. Ordinary least squares provide estimates of coefficients $\beta$.

An individual component $f_{tk}$, for $k = 1, \ldots, K$, can be approximated from the $J$ centered explanatory variables $v_{tjk}$, for $j = 1, \ldots, J$, using

$$\widehat{f}_{tk} = \sum_{j=1}^{J} w_{jk} \phi_{jk} v_{tjk} \tag{4}$$

$$v_{tj1} = x_{tj} - \bar{\mathbf{x}}_j \tag{5}$$

where the centered explanatory variable $v_{tj1}$ for the use in the first component, $k = 1$, is derived by subtracting the respective arithmetic mean, $\bar{\mathbf{x}}_j$, from $x_{tj}$. One derives the specific regression coefficients $\phi_{jk}$ implicitly by regressing the dependent variable on each of the explanatory variables directly:

$$u_{t+h,k} = \sum_{j=1}^{J} \phi_{jk} v_{tjk} \tag{6}$$

$$u_{t+h,1} = y_{t+h} - \bar{\mathbf{y}} \tag{7}$$

where the centered dependent variable $u_{t+h,1}$ for the first component is derived by subtracting the arithmetic mean

of the dependent variable, $\bar{\mathbf{y}}_j$, from the original dependent variable. Furthermore, the algorithm calculates the components iteratively so that the first component is constructed from the original dependent and independent time series. The second component uses only the residuals of the regression of $u_{t+h,1}$ on $f_{t1}$ as the new dependent variable $u_{t+h,2}$ and uses the residuals of the regressions of $v_{tj1}$ on $f_{t1}$ as the new explanatory variables $v_{j2}$. The choice of weights, $w_{jk}$, follows Fuentes et al. (2015) and is set to maximize the covariance between the dependent and independent time series. The solution for the individual weight, $w_{jk}$, for the centered explanatory time series $v_{tjk}$ is:

$$w_{jk} = \frac{\sum_{t=1}^{T} v_{tjk} u_{t+h,k}}{\sqrt{\sum_{j=1}^{J} \left[\sum_{t=1}^{T} v_{tjk} u_{t+h,k}\right]^2}} \qquad \text{for } j = 1, \ldots, J \quad (8)$$

Note that because $v_{tjk}$ and $u_{t+h,k}$ are the centered explanatory and dependent time series, respectively, $\sum_{t=1}^{T} v_{tjk}$ $u_{t+h,k}$ amounts to the covariance of the original, non-centered explanatory and dependent time series times $J$. The collection of all individual weights is the weights vector $w_k = (w_{1k}, \ldots, w_{Jk})'$ of length $J$ as used in Eq. (4).

In contrast to dynamic applications of PLS such as Eickmeier and Ng (2011), this study does not consider the lags of the explanatory variables. In this application, lags of shipping time series are mostly redundant as ships move across the oceans over time. For example, the loads carried by freight ships departing China for the Pacific Ocean today will be observed near the American West Coast in several weeks. Hence, observations of areas near the American West Coast are highly correlated with observations of areas near Asia several weeks ago. Cross-validation determines the optimal number of components $K$ for the specific dependent time series and forecast horizon in line with previous studies (Wold, Ruhe, Wold, & Dunn, 1984; Wold et al., 2001). The maximum number of components is constrained to 10 to avoid overfitting, although rarely more than three components are optimal.

### 2.2. Forecast evaluation

To evaluate the forecast quality of the algorithm described above, the factor model based on PLS calculates expanding window, direct step, and out-of-sample forecasts for the second half of the sample period. The predicted value of the seasonally adjusted monthly growth rate of the trade flow, $y_t$, is denoted by $\widehat{y}^{PLS}_{t+h|t}$. Note that the predicted value changes depending on the time horizon, $h$, with which it was estimated.

Additionally, the PLS forecasts are compared to two benchmark models: A naive model set to the mean growth rate as suggested by Breitung and Knüppel (2021) and a comprehensive model building on Keck et al. (2010), D'Agostino et al. (2017) and Lehmann (2021). The comprehensive model combines a univariate integrated autoregressive-moving average (ARIMA) model and an autoregressive distributed lags (ADL) model with macroeconomic leading indicators. Keck et al. (2010) forecast quarterly trade flows using both a univariate integrated autoregressive-moving average (ARIMA) model, which was initially popularized by Box, Jenkins, Reinsel, and Ljung (1970), as well as a multivariate autoregressive distributed lags (ADL) model with exogenous indicators. Their baseline ARIMA(p,d,q) model specification employs four lags, $p = 4$, one level of differencing, $d = 1$, and moving average terms of length zero, $q = 0$. Their ADL model, in turn, integrates autoregressive terms and quarterly GDP forecasts by the IMF. While the authors find that the ADL model produces superior forecasts overall, the ARIMA model appears to perform well in the near future. Since this study does focus on the near future, the ARIMA model would be the applicable choice. However, D'Agostino et al. (2017) highlight the importance of macroeconomic indicators such as industrial production and purchasing managers indexes, and Lehmann (2021) establishes the forecasting power of survey indicators such as industrial confidence, economic sentiment, and production expectations. Therefore, this study expands the ADL model of Keck et al. (2010) with available macroeconomic indicators and calculates a model combination with an ARIMA model. The individual models to be combined take the following form:

$$y_{t+h} = \alpha_1 + \sum_{i=1}^{p} \beta_i \, y_{t-i} - \sum_{i=1}^{q} \theta_i \, \varepsilon_{t-i} + \eta_{t+h} \quad (9)$$

where $y_{t+h}$ denotes the month-over-month growth rate of the seasonally adjusted time series from period $t+h-1$ to $t+h$. The ARIMA(p,1,q) model with a number of lags, $p$, and a number of moving average (MA) terms, $q$, is optimized using the Akaike Information Criterion (AIC) for every trade flow individually. The MA terms are displayed in the error term representation using $\varepsilon_{t-i}$, while the forecast horizon-specific noise term is denoted by $\eta_{t+h}$. The ADL is enhanced with several leading indicators of the form:

$$y_{t+h} = \alpha_2 + \sum_{i=1}^{2} \beta_i \, y_{t-i} + \sum_{i=0}^{2} \sum_{j=0}^{n} \delta_{ij} \, MACRO_{t-i,j} + \sum_{i=0}^{2} \gamma_i \, SHIP_{t-i} + \sum_{i=0}^{2} \sum_{r=0}^{m} \phi_{ir} \, GLOBAL_{t-i,r} + \varepsilon_{t+h} \quad (10)$$

where *MACRO* includes $n$ macroeconomic indicators such as the European economic sentiment indicators (ESI), industrial confidence indicators (COF), purchasing manager indexes (PMI), and future production surveys (FP), if available for an individual country. Other leading indicators, such as industrial production and order entries, are also selected if the above macroeconomic data series are unavailable for a given country. Table 11 summarizes the selected leading indicators. *SHIP* includes the shipping time series by Cerdeiro et al. (2020) of the IMF transformed to month-over-month growth rates. The $m$ *GLOBAL* indicators, which are identical for all countries, include the New York Federal Reserve's supply chain pressure indicator and the United States industrial production figure. All indicators are seasonally adjusted and converted to monthly growth rates where appropriate and include the contemporaneous data and two lags. The lag structure is

optimized using the AIC. As the PLS inherently forecasts in a direct-step fashion, the ARIMA-ADL combination follows accordingly. The normally distributed error term $\varepsilon_{t+h}$ is unobserved.

The above individual models are combined into one model, labeled the 'benchmark' model:

$$y_{t+h} = \kappa_1 \widehat{y}^{ARIMA}_{t+h|t} + \kappa_2 \widehat{y}^{ADL}_{t+h|t} + \nu_t \tag{11}$$

where $\widehat{y}^{ARIMA}_{t+h|t}$ is the prediction of the ARIMA model, $\widehat{y}^{ADL}_{t+h|t}$ is the respective ADL forecast and $\nu$ is an unobserved error term.

The mean for the naive model and all intercepts and coefficients of the above benchmark models are estimated in an expanding window method in the same fashion as the shipping indicator. Indicators for bilateral trade flows are a union of the indicators for the two trading partners involved. Lastly, some of the aforementioned leading indicators are published with a significant time lag. On September 7, 2022, the latest readings for seaborne cargo, US IP, and port throughput index were for July. In the main results sections, these indicators are lagged by one month. Robustness checks show that lagging indicators by two periods or using indicators without lags does not materially alter the results. The above comprehensive benchmark models do not claim to exhaustively utilize all available information for trade flow forecasts of respective countries. They do represent, however, a natural starting point given the literature on trade forecasts above and include leading indicators that are typically available for a large number of countries.

A standard root mean squared forecast error (RMSFE) calculation evaluates the differences between the out-of-sample forecasts and the dependent time series forecast error, $(\hat{\varepsilon}_{t+h|t})$, across all time periods $t = 1, \ldots, T$ specific to the forecast horizon:

$$RMSFE_h = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\hat{\varepsilon}_{t+h|t}\right)^2} \tag{12}$$

In contrast to other loss functions, such as the mean absolute error, the RMSFE has two distinct advantages: First, it punishes highly for outliers. This project aims to develop a statistical model that accurately traces major shocks to global trade, such as the Corona pandemic. As shocks induce high volatility and forecast errors are likely higher, the RMSFE places high emphasis on these volatile episodes. Second, the RMSFE of the naive model amounts to the standard deviation of the dependent time series. A relative comparison of the factor model RMSFE to the naive model RMSFE can be intuitively interpreted as a comparison to the standard deviation advocated by Breitung and Knüppel (2021). Despite these considerations, the core results are replicated using the mean absolute forecast error as a loss function in the appendix. More formally, a relative RMSFE (rRMSFE) divides the forecast error of the model using the PLS procedure by the (ARIMA-ADL) benchmark model's error.

$$relative\ RMSFE_h = \frac{RMSFE_h^{PLS}}{RMSFE_h^{Bench.}} \tag{13}$$

Hence, a value below 1 for the relative RMSFE indicates lower forecast errors for the new shipping model. Lastly, the Diebold–Mariano test (Diebold & Mariano, 2002) establishes the significance of the difference between the PLS enhanced factor model and benchmark models. Note that more generalized techniques are available to assess the statistical difference between two forecasting models, such as suggested by Giacomini and White (2006). A more generalizable predictive ability test would be necessary if the two models differ in model complexity or estimation procedure, if one model is nested in the other, or if misspecification is likely to occur. Using a component limit of 10 as outlined above, the factor model displays similar complexity to the ARIMA and ADL models estimated with a maximum of 6 and 14 variables, respectively. Both final stages of the models are linear and estimated with ordinary least squares. Neither the results of the PLS calculations nor their inputs are nested in the benchmark models, and the brief period of 46 months in the expanding window out-of-sample procedure leaves little room for model misspecifications due to heterogeneous data and structural breaks. Therefore, the more fundamental Diebold–Mariano may suffice for this project.

Fig. 1 previews results of forecasting imports of individual countries. The figure compares the RMSFE of the PLS forecast on the *y*-axis against the RMSFE of the ARIMA-ADL benchmark forecast on the *x*-axis. A point below the 45°line shows that the PLS procedure produces lower forecast errors than the benchmark model, i.e., the relative RMSFE is below 1.

The above tests may show that the benchmark model outperforms the PLS procedure for some target time series. The PLS algorithm model may hold predictive power for the target time series. This paper constructs an encompassing test following Davidson et al. (1993) to test whether that is the case. First, a model combination, called an encompassing model, combines estimates both from the ARIMA-ADL benchmark model and the estimate of the PLS model:

$$y_{t+h} = \alpha + \sigma_1 \widehat{y}^{ARIMA}_{t+h|t} + \sigma_2 \widehat{y}^{ADL}_{t+h|t} + \sigma_3 \widehat{y}^{PLS}_{t+h|t} + \varepsilon_{t+h} \tag{14}$$

where ordinary least squares estimates coefficients $\sigma_1$, $\sigma_2$ and $\sigma_3$. Let the predictions by the encompassing model be denoted by $\widehat{y}^{Encomp}_{t+h|t}$ and those of the benchmark model by $\widehat{y}^{Bench.}_{t+h|t}$. The encompassing test consists of calculating the Wald statistic for the two null hypotheses

$$H_0^{PLS} : \lambda_1 = 0 \tag{15}$$

$$H_0^{Bench.} : \lambda_2 = 0 \tag{16}$$

where $\lambda_1$ and $\lambda_2$ are implicitly defined in the regression formulae

$$\widehat{y}^{Encomp}_{t+h|t} = \alpha + \lambda_1 \widehat{y}^{PLS}_{t+h|t} + \varepsilon_{t+h} \tag{17}$$

$$\widehat{y}^{Encomp}_{t+h|t} = \alpha + \lambda_2 \widehat{y}^{Bench.}_{t+h|t} + \varepsilon_{t+h}. \tag{18}$$

If the Wald test rejects the null hypothesis $H_0^{PLS} : \lambda_1 = 0$, the PLS is not being encompassed and does hold predictive power beyond competing models. Similarly, if the null hypothesis $H_0^{Bench.} : \lambda_2 = 0$ can be rejected, the benchmark model is not being encompassed and does

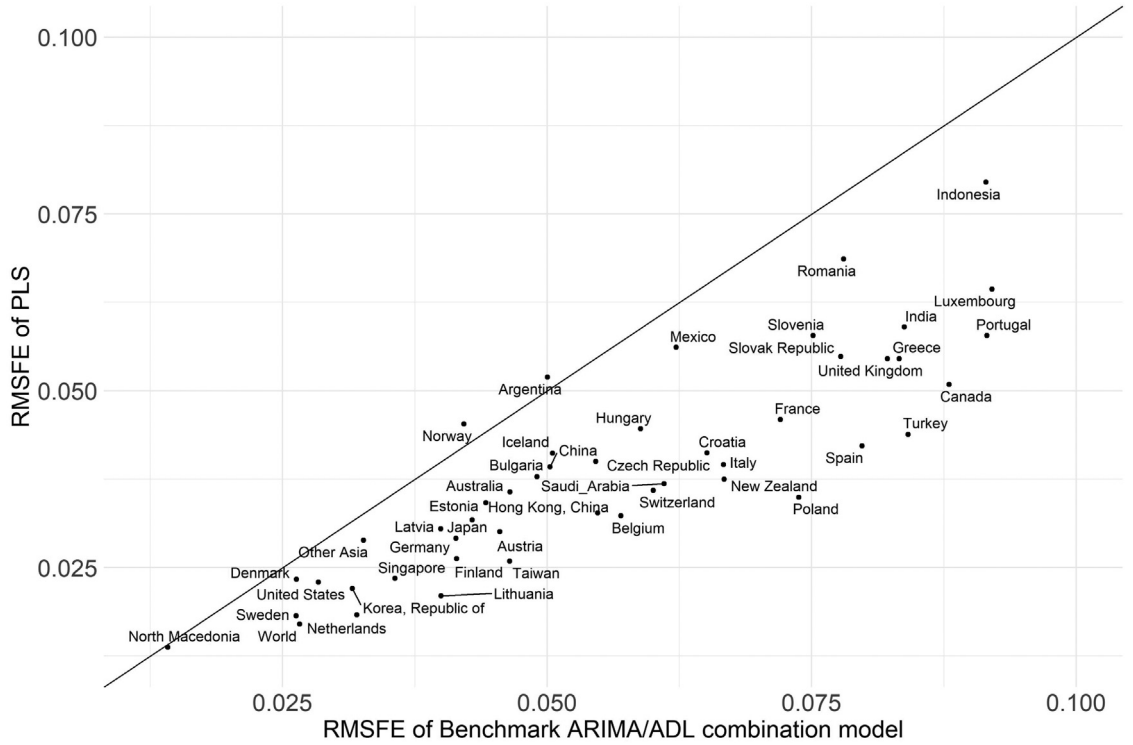



**Fig. 1.** Comparison of forecast errors of PLS shipping model against the benchmark, h = 0.

explain the variation of the target time series beyond the PLS model. Note that both null hypotheses may be rejected, implying that both models hold merit and may be combined. In contrast, one model is considered encompassed by the competing model if the former model's null hypothesis cannot be rejected while that of the competing model is rejected. In other words, the encompassing model solely relies on the competing models while the first is disregarded.

## 3. Data

The data source for container ship locations used in this study is the Automatic Identification System (AIS), a radio system to monitor vessel movements and avoid collisions at sea. Ships broadcast radio signals every few seconds transmitting ship identifiers, course, speed, GPS position, and draught.[1] The maritime intelligence firm FleetMon.com collects the signals from terrestrial receiver stations and satellites interprets the radio signals, and calculates from raw signals the port calls – the stop of cargo ships at a port to load or unload cargo. FleetMon.com provides this project with an exhaustive set of all port calls for container ships over 100 meters in length,[2] as well as one AIS position per day for these container ships. The proprietary data spans the time frame from January 2015 to July 2022 and has global coverage. The 5265 container ships in the sample constitute 99.5% of all ships labeled "container ship" in an independently assembled reference list from classification organizations, including Lloyd's List, society DNV-GL, American Bureau of Shipping, and Nippon Kaiji Kyoka. The following subsections describe the data processing from port calls to time series and AIS positions at sea to time series, respectively.

### 3.1. Predictor time series from port calls

The data on port calls contain approximately 450 thousand observations per year and cover 1346 ports in 173 countries. One observation includes ship identifiers, port identifiers, time and draught of arrival, and time and draught of departure. The data provider calculates this information based on AIS positions with a frequency of several seconds and geographic knowledge of port areas. Fig. 2 validates the port calls at European ports against quarterly maritime transport statistics published by European Commission (2022) for container ships of similar size by port. The figure shows a significant correlation between the two independent data sources. Rotterdam, the largest port in Europe, consistently displays more port calls derived from the AIS system than Eurostat. Since only the correlation of growth rates in the time series is exploited, but not the absolute number of port calls, consistent mismeasurement has no ramifications for the final forecast procedure. Section 5 discusses this further.

To measure the container load of ships, the port call data is combined with technical information for ships

[1] vertical distance between the waterline and the bottom of the ship's hull.

[2] The smallest container ships, so-called feeder ships, which have a carrying capacity of around 300 containers only, typically have a length of 130 m. Hence, the minimum length is not a binding constraint for the vast majority of container ships.

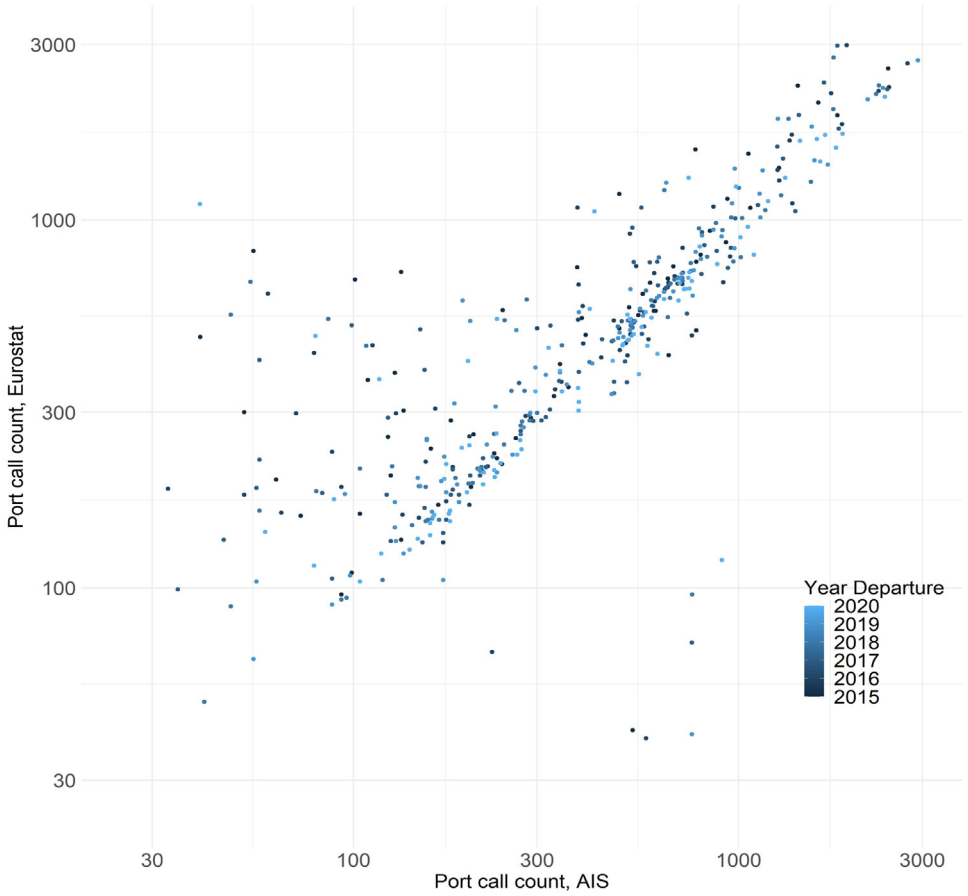



**Fig. 2.** Comparison of number of port calls from Eurostat and AIS.
Notes: Annual number of port calls from Eurostat for container ships larger than 500 Gross Tonnage at the port level (y-axis) and the annual number of port calls for container ships larger than 100 meters in length from AIS data provider Fleetmon.com (x-axis). The shade of the coloring represents the year of observation.

from MarineTraffic, such as minimum draught ("ballast draught"), maximum draught, and capacity measured in Twenty-Foot-Equivalent units (TEU). This technical data assesses the load a ship was carrying when entering and departing a port when it is combined with arrival and departure draughts. A container vessel approaching a port with a draught close to its minimum value likely carries fewer containers than a vessel with a draught near its maximum, ceteris paribus. The current draught is normalized by the potential range of draughts and multiplied by the maximum TEU capacity of the ship to approximate the number of containers the ship currently carries. The following equation translates this into an expression for the TEU load of ship $i$ at time $t$:

$$TEUload_{it} = TEU_i, max \times \frac{draught_{it} - draught_{i,min}}{draught_{i,max} - draught_{i.min}}. \quad (19)$$

This project intentionally does not calculate the net draught change at ports. Since container ships may carry goods towards a port but not unload them, Eq. (19) may consistently overstate the absolute value of the cargo volume unloaded at a port. Since the PLS model uses growth rates of predictor variables, using gross arrival or departure loads does not have ramifications for final predictions as long as the share of cargo (un)loaded is constant over time. In contrast, calculating the net draught change would lead to a complete loss of information when ships unload and load the same amount of cargo. In this case, the net draught change would be zero despite economically meaningful imports and exports.

The approach above reveals several differences to the data preparation steps by Cerdeiro et al. (2020). First, the authors additionally obtain data on dry bulk ships and tankers. Second, the authors apply machine learning tools to calculate the port calls themselves from ship positions but publish the resulting time series directly without using them further in models to forecast trade flows. Consequently, calculating the net-draught change is the more applicable choice for Cerdeiro et al. (2020). This is also because dry bulk ships and tankers typically do not load and unload cargo simultaneously. Furthermore, the gross arrival loads would overestimate goods trade without an additional forecasting tool that maps the time series to trade flows.

To derive time series from this data, the total TEU load of ships entering and departing a port is aggregated at the

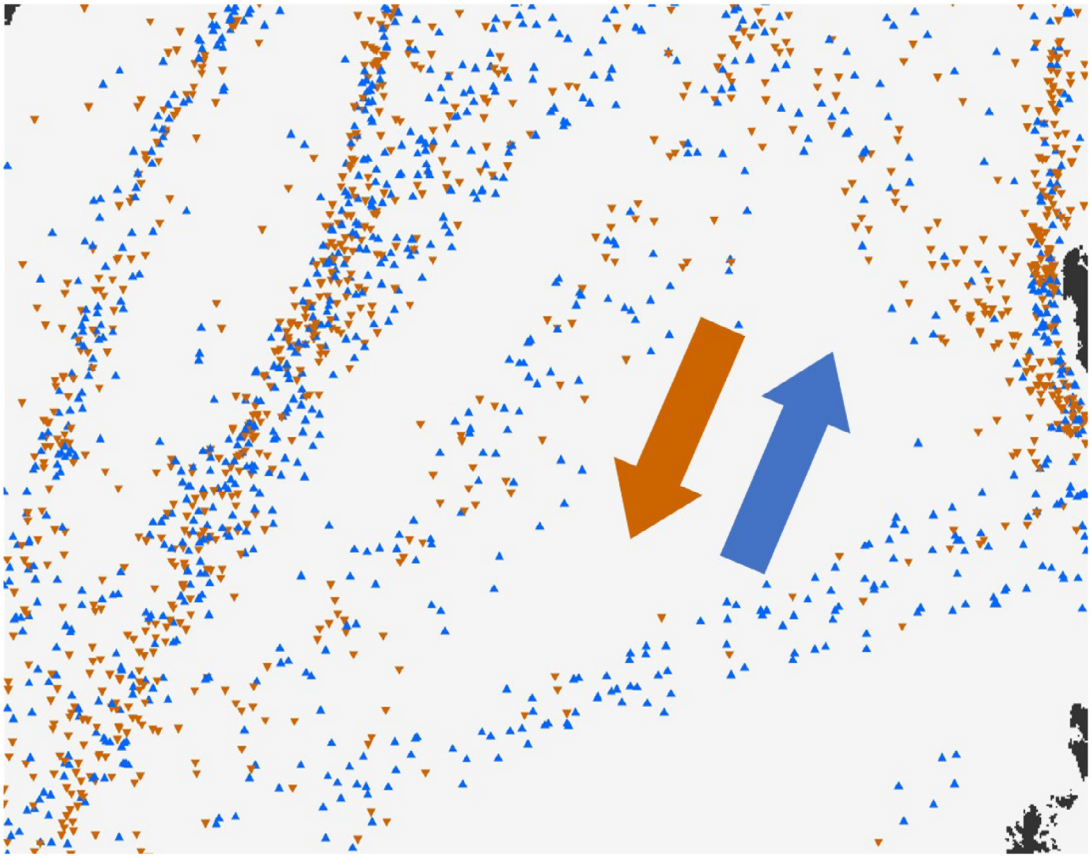


**Fig. 3.** Illustration of ship data aggregation.
The figure shows individual AIS-ship positions in the South China Sea as small triangles. The large arrows illustrate the two predominant courses. For visual purposes, approximately 15% of the image is cropped at the top and bottom.

month and port levels. This yields two time series per port and more than 2600 time series overall: One arrival load time series and one departure load time series per port. The next step is to reduce the number of predictor time series to avoid overfitting the factor model with many independent variables. First, time series that have more than ten monthly observations in the entire period without any reading are omitted. This lowers the number of port call time series to approximately 1200. Second, small ports are added to larger ones in geographic proximity. To that end, the ports are ranked by average monthly TEU load, and the median volume by port, $M$, is calculated. Iteratively, the port with the lowest monthly TEU load is added to the closest port by geographic proximity. This is repeated until no port displays a volume below the original median, $M$. This reduces the number of time series to approximately 680 and significantly reduces the variance of the smallest ports remaining in the dataset. Furthermore, calculating month-over-month growth rates from the above time series derives stationary time series. Using the X-13 procedure of the US Census (US Census Bureau, 2017) on the predictor time series seasonally adjusts the data and reduces cross-correlation in the error terms. This is because clusters of ports in close geographic proximity may be subject to similar seasonal trends through weather conditions, for instance. To avoid ruggedness and increase the flexibility of the setup, the cutoff day for summing monthly data does not have to coincide with the last day of the calendar month. For instance, setting the cutoff day to be the 15th of a month implies summing the data for the 30 days leading up to the 15th. This also generates time series of 12 time periods per year and allows the researcher to produce nowcasts for the current month shortly after the 15th. Except when stated otherwise, the cutoff date in this study is set to be the 30th of the month.

### 3.2. Predictor time series from AIS-positions

To aggregate the positions at sea to economically meaningful predictor time series, the first step partitions the world between 50° latitude South and 70° latitude North into areas of 10° latitude and 10° longitude. The 100 areas with the highest number of ship positions are kept. To give an example, Fig. 3 shows container ship positions for one such area in the South China Sea during July 2020. The color of the arrows indicates a ship's predominant course. Orange triangles represent ships moving on a Southern course, and blue triangles represent ships on a Northern course. Ships going in the opposite direction will likely carry cargo destined for different countries. A k-means clustering algorithm (Hartigan & Wong, 1979) assigns

ships to a predominant course once the course in degrees is mapped into two-dimensional space using the sine and cosine functions.[3] Parallel to the procedure for port calls, single positions are assigned an approximate TEU load by applying Eq. (19). Aggregating the monthly TEU loads of all ships by predominant course yields two time series per area. The resulting 200 series are also seasonally adjusted and can be derived at different cutoff dates during the month.

### 3.3. Other data

Trade data for the dependent time stems from CPB (2022) for price adjusted, unilateral trade flows, from the WTO (2022) for nominal, unilateral trade flows, and from the IMF (2021) for monthly bilateral trade. The ARIMA-ADL benchmark models are calculated using competing leading indicators: The seaborne arrival and departure cargo loads by country are obtained on the IMF website and follow Cerdeiro et al. (2020). The cargo loads will be available beginning in June 2015. European economic sentiment indicators (ESI) and industrial confidence indicators (COF) by country originate from the European Commission (2020). These indicators are matched by country and month to benchmark models. For example, the benchmark model for imports from France includes monthly economic sentiment and industrial confidence indicators for France and the aggregate arriving seaborne cargo load of French ports. Macroeconomic indicators, such as purchasing manager indexes, future production surveys, and others, are obtained from the data platform Thomson Reuters and derived initially from the Directorate-General for Economic and Financial Affairs (DGECFIN), Organization for Economic Co-operation and Development (OECD), companies S&P and Gallup, as well as a wide variety of national sources summarized in Table 11. The US industrial production (US IP) comes from the Federal Reserve Bank of St. Louis (2020) and the global supply chain pressure index (GSCPI) from the Federal Reserve Bank of New York (2022), which are assigned to every country and region. A port throughput index for the Northrange ports (Döhrn & Maatsch, 2012) from RWI (2020) is assigned to the trade flows of Belgium and the Netherlands. Lastly, Table 11 in the appendix lists the leading indicators by country. The benchmark models described above use the seasonally adjusted versions of the indicators if provided and utilize a time series as long as possible.

## 4. Results

### 4.1. Unilateral imports

This section reports the estimation results for unilateral imports. Fig. 4 begins by plotting the nowcasts of the PLS-derived factor model and the benchmark ARIMA-ADL model against actual import growth for six countries and regions, namely the United States, China, the EU, the world in total, Austria, and Sub-Sahara Africa. The black line shows the actual, price, and seasonally adjusted monthly growth of imports, while the gray line shows the out-of-sample PLS predictions for h = 0, i.e., the nowcast. The red line depicts the respective ARIMA-ADL benchmark model. All six panels show a very close fit between dependent variables and the new ship data predictions. This holds true for the landlocked country Austria and the group of developing and emerging countries in Sub-Sahara Africa. Both positive and negative growth rates are estimated well, although the negative COVID-shock to imports in March and April 2020 is captured only in part or not at all. An explanation could be that for the majority of countries, trade of goods by air cargo was earlier and more severely affected by the ramifications of the pandemic as opposed to maritime transport.

The nowcast is available for 76 countries and regions. Table 1 quantifies the results more formally for eleven major economies. In the appendix, Tables 12 and 13 report all results for all countries and regions. The columns labeled RMSFE display the absolute forecast error of the factor model using the PLS procedure (“PLS, Abs”.), The other columns display the relative RMSFE with respect to the naive benchmark model (standard deviation) and the ARIMA-ADL benchmark model, respectively. The data can be interpreted as follows: For German seasonal and price-adjusted imports in the current month (h = 0), the nowcast on the 30th day of the month predicts the month-over-month growth rate with an error of 2.9 percentage points. This amounts to approximately 77.8% of the RMSFE of the naive model (standard deviation) of Brazilian imports and 70.3% of the ARIMA-ADL benchmark model’s error. These relative improvements of the PLS algorithm over the other models are statistically significant as per the Diebold–Mariano test. As comprehensively reported in the Online Appendix Tables 12 and 13, the PLS procedure’s forecast errors in $h = 0$ lie below the standard deviation for 36 countries and regions with statistical significance at the $p < 0.10$ cutoff. For the imports of Italy, Malta, North Macedonia, and Middle-East/North Africa, the forecast error of the benchmark model is smaller than the RMSFE of the model using PLS. The PLS-based model also outperforms the ARIMA-ADL benchmark model with statistical significance for 42 countries. One notable exception is the group of countries in the Middle East/North Africa. Despite the lower forecast errors of the ARIMA-ADL model, the benchmark model does not encompass the forecast based on the PLS procedure. In other words, the new indicator still contributed unique explanatory variation to a combination model.

These results of the encompassing tests are presented in Table 2, complemented by Tables 14 and 15 in the appendix. The column labeled “Wald $p$-value for $H_0^{PLS}$: $\lambda_1 = 0$” reports statistical significance with which the null hypothesis is rejected. The table can be interpreted as follows: The Wald test rejects the null hypothesis for the PLS-derived model of Brazilian imports in h = 0, namely that the PLS procedure is irrelevant for the combination model, with a sufficiently small $p$-value. The ARIMA-ADL benchmark model encompasses the PLS approach

[3] The course in the AIS message is given in degrees from zero to 360, and standard clustering algorithms fail to match the values zero and 360 to the same course without projection into two-dimensional space.

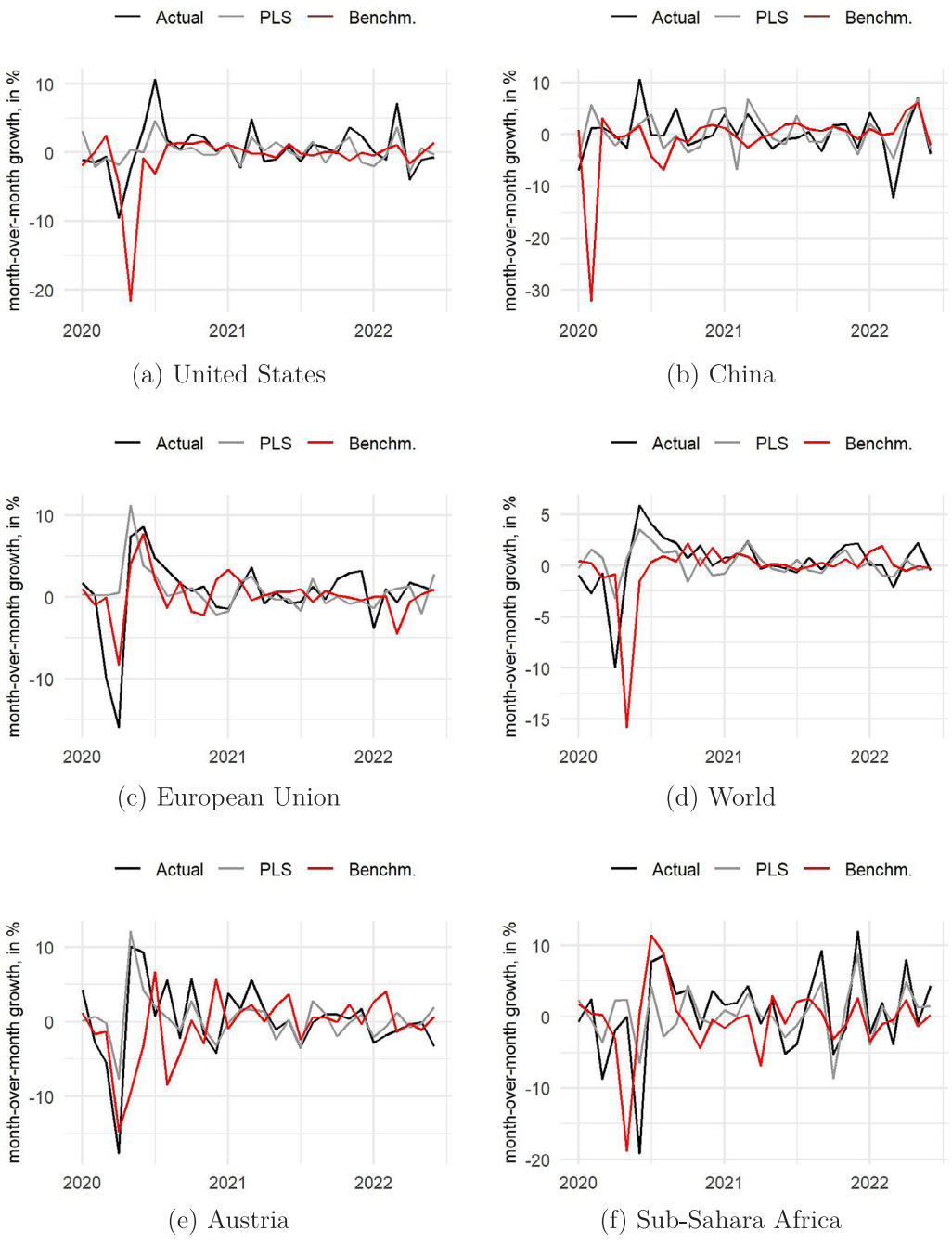



**Fig. 4.** Comparison of nowcast against actual import time series.
The figures show the month-over-month growth rate of the actual import time series and the nowcast from the sparse PLS-procedure.

for four countries and regions: Romania, intra-euro area, intra-European Union, and Middle-East/North Africa. In contrast, the new indicator frequently encompasses the benchmark models. In other words, the null hypothesis can not be rejected that the ARIMA-ADL benchmark model can not contribute explanatory variation to a combination model with the new indicator. As shown in Table 20 in the online appendix, results may hold for a forecast horizon of three months. Forecasts for five months ahead typically do not perform better than the naive mode. Overall, the benchmark model performs relatively poorly. This points to the fact that trade flows are generally difficult to predict. The discussion in Section 5 elaborates on this further and develops robustness checks to improve the performance of the benchmark models.

### 4.2. Exports

Results for unilateral, real exports generally track previous ones for imports. Fig. 5 repeats the visual inspection

**Table 1**
Forecast performance of PLS model for unilateral, real imports, Jan 2015–July 2022.

| | h = 0 | | | h = 1 | | |
|---|---|---|---|---|---|---|
| | RMSFE PLS, Abs. | rel. RMSFE Naive Bench. | rel. RMSFE Bench. | RMSFE PLS, Abs. | rel. RMSFE Naive Bench. | rel. RMSFE Bench. |
| Brazil | 0.069 | 0.925 | 0.651* | 0.059 | 0.639*** | 0.627*** |
| China | 0.039 | 0.934* | 0.780** | 0.034 | 0.819** | 0.606*** |
| EU27 | 0.033 | 0.882 | 0.804 | 0.029 | 0.836 | 0.789 |
| France | 0.045 | 1.037 | 0.637* | 0.046 | 0.985 | 0.740** |
| Germany | 0.029 | 0.778* | 0.703** | 0.026 | 0.729* | 0.620*** |
| Japan | 0.031 | 1.033 | 0.739** | 0.027 | 0.898 | 0.780** |
| Russia | 0.019 | 0.745** | 0.860 | 0.016 | 0.721** | 0.732* |
| Sub-S. Afr. | 0.037 | 0.752** | 0.712* | 0.038 | 0.811* | 0.699 |
| UK | 0.054 | 0.822* | 0.663*** | 0.049 | 0.686* | 0.523*** |
| USA | 0.022 | 0.752* | 0.808** | 0.025 | 0.808* | 0.638*** |
| World | 0.017 | 0.740 | 0.639** | 0.023 | 0.754 | 0.705 |

RMSFE refers to root mean squared forecast error. PLS, Abs. denotes the absolute RMSFE of the PLS model in decimal units. Rel. RMSFE Naive Bench. and rel. RMSFE Benchm. denote the relative RMSFEs, see Eq. (13), of the sparse model using the PLS procedure against the naive and ARIMA-ADL benchmark models, respectively. A value below 1 indicates lower forecast errors of the PLS model. Relative RMSFE values above one marked in gray. Benchmark models are ARIMA-ADL models using economic indicators applicable to the country. Significance levels calculated using Diebold-Mariano test. Sub-S. Afr. designates Sub-Saharan Africa. Cutoff day for PLS = 30.
*p<0.1; **p<0.05; ***p<0.01

**Table 2**
Encompassing tests of PLS and benchmark models for unilateral, real imports, Jan 2015–July 2022.

| | h = 0 | | h = 1 | |
|---|---|---|---|---|
| | Wald p-Value $H_0^{PLS}$: $\lambda_1 = 0$ | Wald p-Value $H_0^{Bench.}$: $\lambda_2 = 0$ | Wald p-Value $H_0^{PLS}$: $\lambda_1 = 0$ | Wald p-Value $H_0^{Bench.}$: $\lambda_2 = 0$ |
| Brazil | 0.011** | 0.303 | 0.000*** | 0.016** |
| China | 0.002*** | 0.778 | 0.000*** | 0.167 |
| EU27 | 0.004*** | 0.133 | 0.000*** | 0.000*** |
| France | 0.011** | 0.035** | 0.031** | 0.000*** |
| Germany | 0.000*** | 0.220 | 0.000*** | 0.470 |
| Japan | 0.166 | 0.155 | 0.014** | 0.767 |
| Russia | 0.000*** | 0.001*** | 0.000*** | 0.019** |
| Sub-S. Afr. | 0.000*** | 0.033** | 0.000*** | 0.617 |
| UK | 0.000*** | 0.545 | 0.000*** | 0.778 |
| USA | 0.000*** | 0.280 | 0.000*** | 0.534 |
| World | 0.000*** | 0.595 | 0.000*** | 0.460 |

Significance of the Wald test implies rejection of the null hypothesis and indicates not encompassed predictive power of the underlying model. See equations Eq. (15) and following for definitions. Cells in gray show encompassed models: They do not explain variation of the combination model, while the competing model does. Cutoff day for PLS = 30.
*p<0.1; **p<0.05; ***p<0.01.

of forecasts for six sample countries and regions. The shipping model works surprisingly well for the landlocked country of Austria. This shows the model's strength in exploiting correlations between German and Italian ports and global trends with Austrian exports.

Table 3 shows that the results for export forecasts generally match those for imports, although forecast errors for exports are, on average, one percentage point larger. For instance, errors for German imports in $h = 0$ are 2.9 pp, while those for German exports are 4.9 pp. Additionally, the forecast errors for exports tend to increase more strongly as one predicts with a forecast horizon of $h = 1$. Tables 16 and 17 report the results for all countries and regions. For 31 countries and regions, the PLS-based indicator produces significantly lower forecast errors than the naive model. This number increases to 43 countries and regions for the comparison against the main benchmark model. The encompassing tests of Tables 4, as well as Tables 18 and 19 show that the PLS procedure for unilateral exports in the current and subsequent month is rarely encompassed, but frequently encompasses the benchmark model.

Again, Table 22 in the Online Appendix shows results for the forecast horizons of three and five months. Using a five-month forecast horizon renders predictions by the sparse PLS procedure indistinguishable from a naive model for all ten selected countries. Additionally, the forecasts for exports appear to worsen more quickly than those for imports as the forecast horizon increases. The economic explanation is that imported goods from a country are also frequently transported away from the origin country several months before arrival at the destination country. Export goods are typically transported only within a country and on land before the products

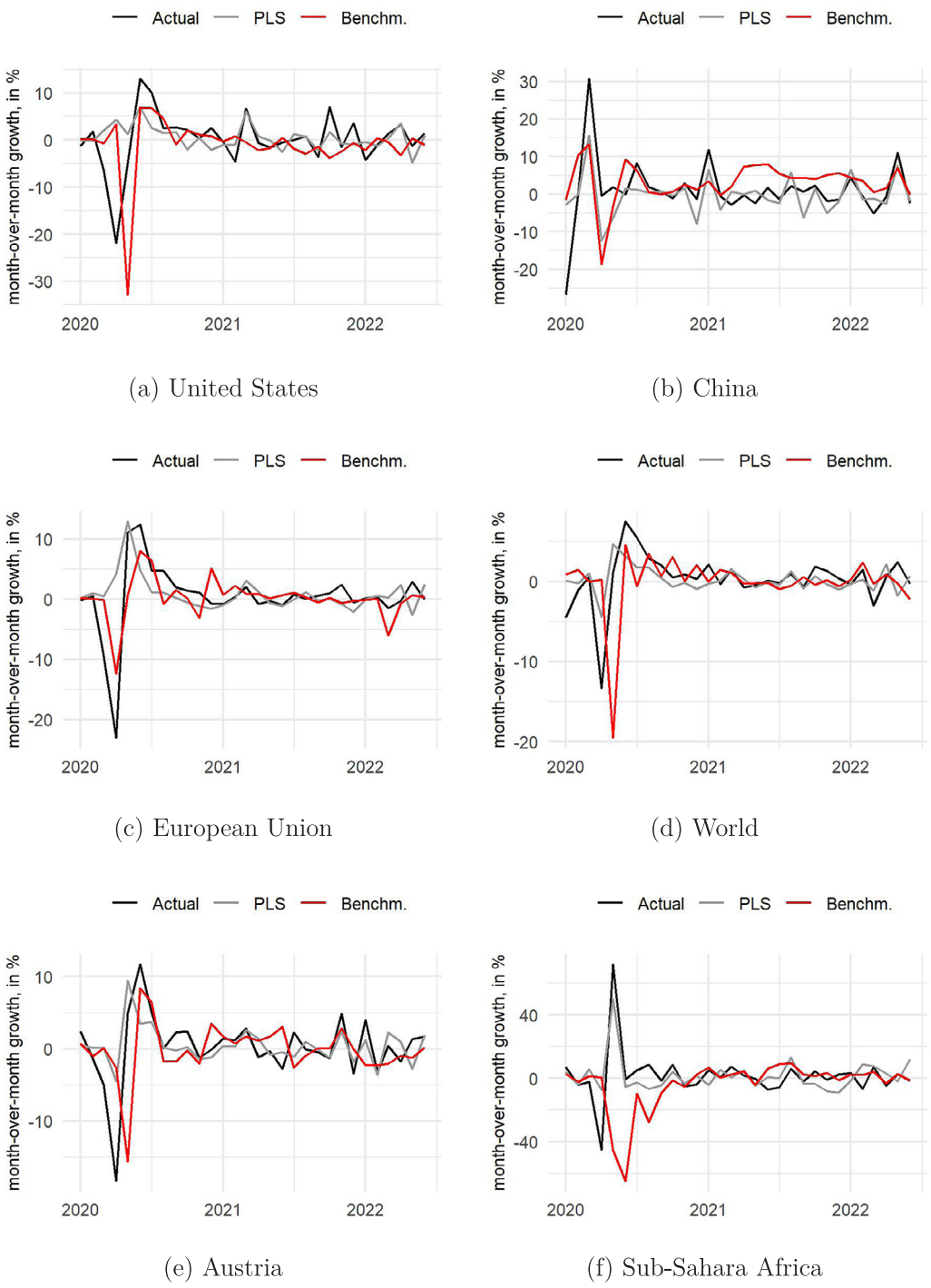



**Fig. 5.** Comparison of nowcast against actual export time series.
The figures show the month-over-month growth rate of the actual export time series and the nowcast from the sparse PLS model.

leave the country. Hence, the container shipping network observes imports several months earlier.

### 4.3. Bilateral trade

The flexible setup of the PLS algorithm also allows forecasting of specific bilateral trade flows. To exemplify this, the algorithms are trained on the bilateral imports of China, the European Union, and the United States vis-à-vis their five largest trade partners. Table 5 reports the absolute and relative RMSFE values. The relative RMSFE against the naive model is remarkably similar to the performance of the forecasts for unilateral imports. For instance, the RMSFEs of the shipping model for the EU's

**Table 3**
Forecast performance of PLS model for unilateral, real exports, Jan 2015–July 2022.

| | h = 0 | | | h = 1 | | |
|---|---|---|---|---|---|---|
| | RMSFE PLS, Abs. | rel. RMSFE Naive Bench. | rel. RMSFE Bench. | RMSFE PLS, Abs. | rel. RMSFE Naive Bench. | rel. RMSFE Bench. |
| Brazil | 0.049 | 0.575*** | 0.590*** | 0.052 | 0.679*** | 0.617*** |
| China | 0.058 | 0.946 | 0.751* | 0.059 | 1.185 | 0.838 |
| EU27 | 0.048 | 1.031 | 1.150 | 0.049 | 1.166 | 0.969 |
| France | 0.057 | 1.054 | 0.510 | 0.061 | 0.978 | 0.856 |
| Germany | 0.049 | 1.096 | 0.941 | 0.048 | 1.033 | 0.993 |
| Japan | 0.028 | 0.797* | 0.841** | 0.032 | 0.689*** | 0.638** |
| Russia | 0.035 | 0.941 | 1.314** | 0.041 | 0.962 | 0.769** |
| Sub-S. Afr. | 0.090 | 1.020 | 0.660 | 0.096 | 0.908 | 0.623 |
| UK | 0.057 | 0.755* | 0.747*** | 0.046 | 0.587** | 0.566*** |
| USA | 0.049 | 0.739 | 0.730** | 0.045 | 0.711 | 0.613*** |
| World | 0.021 | 0.904 | 0.735* | 0.026 | 0.950 | 0.956 |

RMSFE refers to root mean squared forecast error. PLS, Abs. denotes the absolute RMSFE of the PLS model in decimal units. Rel. RMSFE Naive Bench. and rel. RMSFE Benchm. denote the relative RMSFEs, see Eq. (13), of the sparse model using the PLS procedure against the naive and ARIMA-ADL benchmark models, respectively. A value below 1 indicates lower forecast errors of the PLS model. Relative RMSFE values above one marked in gray. Benchmark models are ARIMA-ADL models using economic indicators applicable to the country. Significance levels calculated using Diebold-Mariano test. Sub-S. Afr. designates Sub-Saharan Africa. Cutoff day for PLS = 30.
*p<0.1; **p<0.05; ***p<0.01.

**Table 4**
Encompassing tests of PLS and benchmark models for unilateral, real exports, Jan 2015–July 2022.

| | h = 0 | | h = 1 | |
|---|---|---|---|---|
| | Wald p-Value $H_0^{PLS}$: $\lambda_1 = 0$ | Wald p-Value $H_0^{Bench.}$: $\lambda_2 = 0$ | Wald p-Value $H_0^{PLS}$: $\lambda_1 = 0$ | Wald p-Value $H_0^{Bench.}$: $\lambda_2 = 0$ |
| Brazil | 0.000*** | 0.435 | 0.000*** | 0.301 |
| China | 0.000*** | 0.246 | 0.422 | 0.969 |
| EU27 | 0.032** | 0.000*** | 0.012** | 0.205 |
| France | 0.016** | 0.003*** | 0.152 | 0.001*** |
| Germany | 0.037** | 0.017** | 0.397 | 0.021** |
| Japan | 0.000*** | 0.607 | 0.000*** | 0.209 |
| Russia | 0.001*** | 0.001*** | 0.659 | 0.094* |
| Sub-S. Afr. | 0.000*** | 0.018** | 0.000*** | 0.002*** |
| UK | 0.001*** | 0.639 | 0.000*** | 0.152 |
| USA | 0.000*** | 0.015** | 0.000*** | 0.342 |
| World | 0.001*** | 0.378 | 0.000*** | 0.005*** |

Significance of the Wald test implies rejection of the null hypothesis and indicates not encompassed predictive power of the underlying model. See equations Eq. (15) and following for definitions. Cells in gray show encompassed models: They do not explain variation of the combination model, while the competing model does. Sub-S. Afr. designates Sub-Saharan Africa. Cutoff day for PLS = 30.
*p<0.1; **p<0.05; ***p<0.01.

imports from its top trade partners range from 75% to 96% of the RMSFE of the naive model (standard deviation). This compares to a relative performance of 88% against the naive model for imports of the EU in total. Since the baseline standard deviation is much higher for bilateral trade flows, the absolute forecast errors for bilateral trade flows are also higher. As shown in Table 6, the forecasts for bilateral trade flows are never encompassed and, in turn, frequently encompass the benchmark models.

### 4.4. Performance during crisis

Largely unexpected shocks such as the COVID-19 crisis make it more difficult to pinpoint the magnitude of disruptions to trade flows. To study the properties of the shipping indicator in times of crisis, this subsection compares the PLS model forecasts during the first half of 2020 and the preceding two years, during which there were no COVID-related lockdowns. The figure consists of three panels: The top panel shows the relative RMSFE of the PLS-derived indicator against the pure ARIMA model of Eq. (9). The second panel displays the comparison against a model derived from the ADL model of Eq. (10) but omits the autoregressive terms. The bottom panel shows the comparison against the shipping indicator calculated by Cerdeiro et al. (2020). This split between the elements of the benchmark model attempts to illustrate the relative performance of the new PLS indicator in times of crises compared to autoregressive

**Table 5**
Forecast performance of PLS model for bilateral trade, Jan 2015–Jul 2022.

| | h = 0 | | | h = 1 | | |
|---|---|---|---|---|---|---|
| | RMSFE PLS, Abs. | rel. RMSFE Naive | rel. RMSFE Bench. | RMSFE PLS, Abs. | rel. RMSFE Naive | rel. RMSFE Bench. |
| CN from AU | 0.125 | 1.015 | 0.929 | 0.077 | 0.691** | 0.628* |
| CN from DE | 0.051 | 0.865 | 0.751* | 0.053 | 0.837 | 0.605*** |
| CN from JP | 0.046 | 0.924 | 0.959 | 0.040 | 0.832 | 0.738** |
| CN from KR | 0.038 | 0.674*** | 0.661*** | 0.040 | 0.759*** | 0.764** |
| CN from US | 0.048 | 0.724*** | 0.715** | 0.043 | 0.684** | 0.636** |
| EU from CN | 0.047 | 0.763 | 0.573 | 0.045 | 0.703 | 0.483* |
| EU from RU | 0.075 | 0.912 | 1.093 | 0.083 | 0.989 | 0.956 |
| EU from CH. | 0.029 | 0.827 | 0.600** | 0.036 | 0.972 | 0.755 |
| EU from UK | 0.102 | 0.962 | 0.718* | 0.094 | 0.805 | 0.716 |
| EU from US | 0.046 | 0.754** | 0.770** | 0.046 | 0.688*** | 0.633*** |
| US from CA | 0.061 | 0.924 | 0.656** | 0.077 | 1.007** | 0.787 |
| US from CN | 0.069 | 1.197 | 0.833 | 0.064 | 1.056 | 0.646 |
| US from DE | 0.052 | 0.679** | 0.787** | 0.054 | 0.678** | 0.629** |
| US from JP | 0.056 | 0.816** | 0.799* | 0.050 | 0.649** | 0.617** |
| US from MX | 0.105 | 1.138 | 0.820 | 0.103 | 1.330 | 0.871 |

RMSFE refers to root mean squared forecast error. PLS, Abs. denotes the absolute RMSFE of the PLS model in decimal units. Rel. RMSFE Naive and rel. RMSFE Benchmark denote the relative RMSFEs of the PLS model divided by the standard deviation and the RMSFE of the benchmark model, respectively. A value below 1 indicates lower forecast errors of the PLS model. Relative RMSFE values above one marked in gray. Significance levels calculated using Diebold-Mariano test. Cutoff day for PLS = 30.
*p<0.1; **p<0.05; ***p<0.01.

**Table 6**
Encompassing tests of PLS and benchmark models for bilateral trade, Jan 2015–Jul 2022.

| | h = 0 | | h = 1 | |
|---|---|---|---|---|
| | Wald p-Value $H_0^{PLS}$: $\lambda_1 = 0$ | Wald p-Value $H_0^{Bench.}$: $\lambda_2 = 0$ | Wald p-Value $H_0^{PLS}$: $\lambda_1$=0 | p-Value $H_0^{Bench.}$: $\lambda_2 = 0$ |
| CN from AU | 0.003*** | 0.760 | 0.000*** | 0.102 |
| CN from DE | 0.000*** | 0.644 | 0.000*** | 0.683 |
| CN from JP | 0.015** | 0.024** | 0.000*** | 0.152 |
| CN from KR | 0.000*** | 0.255 | 0.000*** | 0.775 |
| CN from US | 0.000*** | 0.519 | 0.000*** | 0.336 |
| EU from CN | 0.000*** | 0.432 | 0.000*** | 0.188 |
| EU from RU | 0.005*** | 0.000*** | 0.005*** | 0.018** |
| EU from CH | 0.000*** | 0.356 | 0.000*** | 0.024** |
| EU from UK | 0.000*** | 0.763 | 0.000*** | 0.873 |
| EU from US | 0.000*** | 0.002*** | 0.000*** | 0.141 |
| US from CA | 0.000*** | 0.085* | 0.002*** | 0.009*** |
| US from CN | 0.000*** | 0.142 | 0.000*** | 0.053* |
| US from DE | 0.000*** | 0.614 | 0.000*** | 0.931 |
| US from JP | 0.000*** | 0.052* | 0.000*** | 0.155 |
| US from MX | 0.000*** | 0.005*** | 0.000*** | 0.301 |

Significance of the Wald test implies rejection of the null hypothesis and indicates not encompassed predictive power of the underlying model. See equations Eq. (15) and previous for definitions. Cells in gray show encompassed models: They do not explain variation of the combination model, while the competing model does. Cutoff day for PLS = 30.
*p<0.1; **p<0.05; ***p<0.01.

and non-autoregressive terms. All panels show the relative RMSFE for the first half of 2020 in dark gray and the period from 2015 to 2019 in light gray. The first panel shows that the PLS-derived indicator improves the relative performance against the ARIMA model during the first half of 2020. The chart shows this particularly well for the United Kingdom, France, and the world. The picture is more mixed for the comparison against the macroeconomic indicators. The macroeconomic indicators for the EU and Germany outperformed the PLS indicator in the first half of 2020. Lastly, the bottom panel illustrates that the new PLS model outperforms the raw shipping indicator without a forecasting model.

### *4.5. Performance against other selection procedures and machine learning algorithms*

The objective of the above analyses is to show that the sparse PLS model generates estimates that are useful for forecasting in absolute terms and relative to benchmark models. A secondary question remains whether the sparse PLS model is the best machine learning (ML) tool to estimate trade flows with the AIS data. Similarly, one

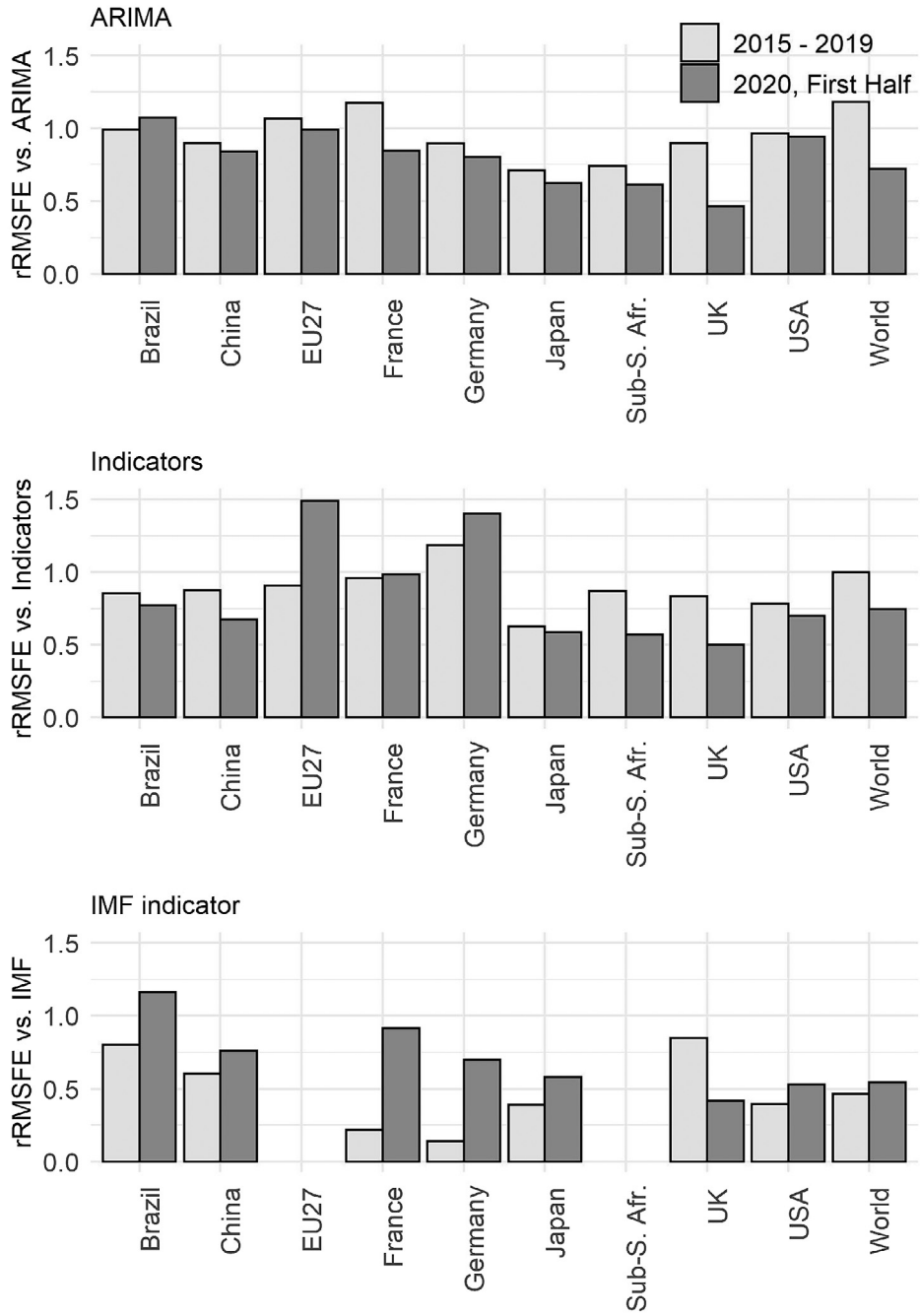



**Fig. 6.** Relative RMSFE of new shipping indicator (PLS) vs. ARIMA-ADL-model before and during COVID-19 crisis.

may question if the regular least absolute shrinkage and selection operator (LASSO) is the most applicable to perform the preselection of variables. First, this subsection explores the performance of sparse PLS against other estimation methods. Specifically, LASSO as a stand-alone estimator, a regression tree using cross-validation methods for "pruning" – see, Januschowski et al. (2021) for an overview of associated methods –, a random forest, a principle component regression (PCR) and a PLS system are trained on shipping data with and without shrinkage by LASSO. The results are compared to the PLS model with shrinkage as used above using the relative RMSFE indication:

$$relative\ RMSFE_h = \frac{RMSFE_h^{SparsePLS}}{RMSFE_h^{Other\ ML}} \tag{20}$$

Table 7 reports the results for export predictions of the different estimation techniques.

The first column "Sparse PLS, abs. RMSFE" lists the absolute forecast error of the out-of-sample forecasts of

**Table 7**
Comparison of various machine learning tools with and without sparcity against the sparse PLS; real exports, Jan 2015–July 2022.

| | Sparse PLS abs.RMSFE | Lasso rRMSFE | PCR rRMSFE | Reg. Tree rRMSFE | R. Forest rRMSFE | Sparse Lasso rRMSFE | Sparse PCR rRMSFE | Sparse Tree rRMSFE |
|---|---|---|---|---|---|---|---|---|
| h=0 | | | | | | | | |
| Brazil | 0.042 | 0.561** | 0.570*** | 0.523*** | 0.594*** | 0.734** | 1.015 | 0.578*** |
| China | 0.043 | 0.590* | 0.549** | 0.530** | 0.575** | 0.625** | 1.006 | 0.567** |
| EU27 | 0.036 | 0.774* | 0.736** | 0.750** | 0.769* | 0.780** | 0.901 | 0.713** |
| France | 0.045 | 0.713* | 0.677** | 0.651** | 0.710* | 0.728* | 1.267 | 0.658** |
| Germany | 0.036 | 0.697** | 0.654** | 0.655** | 0.688** | 0.770 | 1.363 | 0.691*** |
| Japan | 0.027 | 0.676** | 0.636*** | 0.656** | 0.671** | 0.683** | 1.308*** | 0.629*** |
| Russia | 0.032 | 0.780 | 0.793 | 0.777* | 0.767 | 0.783 | 0.891 | 0.815 |
| Sub-S. Afr. | 0.081 | 0.577** | 0.583 | 0.558 | 0.606 | 0.679** | 1.209 | 0.582 |
| UK | 0.042 | 0.602*** | 0.593*** | 0.554*** | 0.603*** | 0.623*** | 0.991 | 0.571*** |
| USA | 0.038 | 0.779** | 0.724** | 0.753** | 0.757** | 0.783** | 0.930 | 0.765** |
| World | 0.019 | 0.702* | 0.667** | 0.644** | 0.695* | 0.706 | 1.017 | 0.677*** |
| h=1 | | | | | | | | |
| Brazil | 0.049 | 0.648*** | 0.687*** | 0.617*** | 0.687*** | 0.736*** | 0.985 | 0.630*** |
| China | 0.037 | 0.524** | 0.515** | 0.441*** | 0.521** | 0.547** | 0.945 | 0.501*** |
| EU27 | 0.042 | 0.886 | 0.775 | 0.823 | 0.877 | 0.914 | 1.071 | 0.884 |
| France | 0.060 | 0.963 | 0.852 | 0.862 | 0.901* | 0.939 | 1.279 | 0.905* |
| Germany | 0.045 | 0.897 | 0.764 | 0.778 | 0.845 | 0.890 | 1.103 | 0.789 |
| Japan | 0.030 | 0.729*** | 0.732*** | 0.747*** | 0.745*** | 0.751*** | 1.017 | 0.756*** |
| Russia | 0.033 | 0.801 | 0.809 | 0.815 | 0.785 | 0.801 | 1.032 | 0.846 |
| Sub-S. Afr. | 0.087 | 0.636 | 0.618 | 0.571 | 0.648 | 0.641 | 1.088 | 0.646 |
| UK | 0.038 | 0.555*** | 0.518*** | 0.509*** | 0.547** | 0.580*** | 0.984 | 0.546*** |
| USA | 0.043 | 0.890** | 0.871 | 0.836 | 0.883*** | 0.892** | 0.972 | 0.784** |
| World | 0.016 | 0.592* | 0.558* | 0.603* | 0.576** | 0.593* | 1.101* | 0.636** |

Note: The column "Sparse PLS, abs. RMSFE" measures the absolute forecast error of the sparse partial least squares model. Columns labeled "rRMSFE" measure the relative RMSFE of the sparse PLS as compared to the noted machine learning tools. "Sparse" always refers to prior shrinkage of the predictor space using LASSO.

the sparse PLS-derived factor model as used previously. The subsequent columns measure the relative forecast error compared to the competing ML tools. A first insight is that the sparse PLS approach delivers significantly lower forecast errors than the machine learning tools without preselection. The relative RMSFEs against the stand-alone LASSO, the PCR, regression tree, and random forest range between approximately 0.5 and 0.8 for the nowcast. Among the competing models without preselection, the random forest achieves the highest relative RMSFEs with 0.769 for EU-27 exports in the current month. The last three columns show the performance of the sparse PLS procedure against sparse LASSO, sparse PCR, and sparse regression tree methods. The sparse PCR tool produces forecasts with very similar accuracy to the sparse PLS technique. The sparse PCR yields significantly lower forecast errors for Japanese export nowcasts, as indicated by the RMSFE being above one. The good performance of sparse PCR against sparse PLS is not surprising as the two factoring procedures are closely related. The most important difference between the techniques is that PLS considers the covariance between predictor and dependent variables when forming the components, while PCR only considers the covariance between predictors. However, this difference is masked by LASSO, which also considers the correlation between predictor and dependent time series. Of course, the above list of machine learning tools is not exhaustive. Employing Bayesian estimation methods and applying neural networks may produce different results. Moreover, there are several natural alternatives to the variable preselection besides the standard LASSO approach. Notably, the following tests the performance of the standard LASSO approach as a selection tool against the elastic net (Zou & Hastie, 2005), which combines penalties of the ridge and lasso regression, the adaptive LASSO (Zou, 2006), a stepwise (forward) selection operator based on ordinary least squares estimation and the Bayesian information criterion (BIC), as well as the variable importance on projection (VIP). The variable influence on projection (VIP) is native to the partial least squares technique and has gained traction in the chemometrics literature (Cocchi, Biancolillo, & Marini, 2018; Galindo-Prieto, Eriksson, & Trygg, 2014; Tran, Afanador, Buydens, & Blanchet, 2014). The statistic can be calculated for predictor time series $j$ as such:

$$VIP_j = \sqrt{K \frac{\sum_{k=1}^{K} SSY_k \left(\frac{w_{kj}}{\|w_k\|}\right)^2}{SSY_{total}}} \tag{21}$$

where $SSY_k$ is the sum of squares explained by individual component k, $SSY_{total}$ is the sum of squares explained by the entire factor model, and $K$ is the total number of factors. The sums of squares explained by components are multiplied by the weights of the individual predictor variables to reflect the predictor's influence on the dependent time series through the various components. VIP identifies the 35–45 most important explanatory variables. Results are shown in Table 8. The PLS technique produces reliably lower forecast errors when shrinkage is performed with regular LASSO instead of the stepwise selection or the VIP tool—the relative RMSFEs range between 0.71 and 0.95. On the contrary, comparing the regular LASSO with an elastic net or an adaptive PLS

**Table 8**
Comparison of various selection tools and PLS procedure against regular LASSO and PLS procedure; real exports, Jan 2015–July 2022.

| | Lasso abs.RMSFE | Adaptive Lasso rRMSFE | Elastic Net rRMSFE | Stepwise OLS rRMSFE | VIP rRMSFE |
|---|---|---|---|---|---|
| h=0 | | | | | |
| Brazil | 0.042 | 0.849 | 0.862* | 0.746* | 0.814* |
| China | 0.043 | 0.916 | 1.005 | 0.721*** | 0.711** |
| EU27 | 0.036 | 0.922** | 1.000 | 0.843 | 0.739 |
| France | 0.045 | 0.819 | 1.014 | 0.914 | 0.779 |
| Germany | 0.036 | 0.909 | 1.028 | 0.908** | 0.827* |
| Japan | 0.027 | 1.027 | 1.045** | 0.722*** | 0.942 |
| Russia | 0.032 | 0.962 | 1.007 | 0.872 | 0.879** |
| Sub-S. Afr. | 0.081 | 0.985 | 0.992 | 0.773** | 0.821* |
| UK | 0.042 | 0.841*** | 0.995 | 0.721*** | 0.737*** |
| USA | 0.038 | 0.856 | 1.000 | 0.842*** | 0.743 |
| World | 0.019 | 1.085 | 1.017 | 0.815** | 0.911** |
| h=1 | | | | | |
| Brazil | 0.049 | 1.065 | 1.025 | 0.716** | 0.971 |
| China | 0.037 | 1.042 | 0.897*** | 0.681 | 0.601** |
| EU27 | 0.042 | 0.949 | 0.994 | 0.907* | 0.925 |
| France | 0.060 | 0.984 | 0.997 | 0.940 | 0.995 |
| Germany | 0.045 | 0.974 | 1.011 | 0.892* | 0.936 |
| Japan | 0.030 | 0.888 | 1.017 | 0.815*** | 0.944 |
| Russia | 0.033 | 0.936 | 0.988 | 0.874* | 0.837 |
| Sub-S. Afr. | 0.087 | 0.996 | 1.050 | 0.844 | 0.881* |
| UK | 0.038 | 0.926 | 1.089 | 0.722** | 0.786*** |
| USA | 0.043 | 1.040* | 1.000 | 0.940 | 1.003 |
| World | 0.016 | 0.914 | 0.974 | 1.963 | 0.666*** |

Note: The column "Lasso, abs. RMSFE" measures the absolute forecast error of the sparse partial least squares model using the regular LASSO procedure. Columns labeled "rRMSFE" measure the relative RMSFE of the PLS procedure with regular LASSO as compared to the noted selection tools and PLS.

technique shows that forecast errors are indistinguishable. As for the nowcast, preselection with the elastic net yields slightly lower forecast errors for five countries in the displayed group of countries, while regular LASSO yields lower or identical errors for the remaining five countries. Only two results are significant, as per the Diebold–Mariano test. Perfectly identical results occur when the elastic net and regular LASSO select the same 45 time series. As a result, the forecast errors of shrinkage with regular LASSO, adaptive LASSO, and elastic net indicate that the three methods are interchangeable for this application.

## 5. Discussion and robustness

### 5.1. Effects of measurement error and structural breaks

Data preparation of the AIS data plays an important part in this study. Hence, this subsection discusses the effect potential measurement errors in the shipping data have on the final forecasts. The first type of potential measurement error concerns an inaccurate approximation of the level of the AIS-derived time series. One may over- or understate the economic activity level at ports or sea areas if, for instance, ships are incorrectly classified as "container ships". The maritime intelligence firm, which provides the AIS data for this study, may also define specific ports differently than other sources. For instance, what is commonly referred to as the "Port of Singapore" is, in fact, a group of ports with over a dozen terminals. Such variance in classifications may explain why the port of Rotterdam has more AIS port calls per month than recorded by Eurostat in Fig. 2. Other measurement error sources at the cargo load level include the seawater's salinity and temperature and their distorting effect on the ships' draught or the transport of containers into a port without unloading them. As long as the distortions are constant over time, monthly growth rates will mask the measurement errors, and final forecasts will not change.

A second form of potential measurement error concerns the growth rate of the predictor time series. For example, an underestimation of the ballast draught, i.e., minimum draught, or a non-linear relationship between the draught and the cargo volume due to the form of a ship's hull could lead to an attenuated growth rate in comparison to the actual growth rate of cargo carried. Suppose such measurement error of the growth rate is constant over time. The PLS model will correct this by adjusting the variable regression coefficient $\phi$ in Eq. (4) according to the observed correlation between independent and dependent time series. A similar argument holds for other related trends that change the correlation between dependent and independent time series: Transshipment of cargo implies that not all containers discharged at a port are destined to be imported to the port's country. Instead, a second vessel likely carries them to a different country. The Port of Singapore constitutes a well-known example of this. Similarly, many containers unloaded at the port of Rotterdam in the Netherlands are transported by trucks to Germany. These phenomena do not invalidate the PLS forecasts: Both the LASSO and PLS systems exploit the empirically observed correlation in the training phase

between port cargo flows and country trade flows and adjust the forecast model accordingly.

A more serious threat to the forecasting models is structural breaks in the data, which can result from measurement errors or natural occurrences. A one-time change of satellite coverage or manipulation of AIS data could induce such a structural break in the level of the predictor time series. Due to the conversion of time series to growth rates, this structural break only causes the growth rate to display one aberrant observation. The correlation between independent and dependent time series will diminish if this break occurs in the training phase. As a result, the time series may not be selected for the final forecast model. However, if this structural break occurs in the testing phase and the predictor time series is selected for the final model, one out-of-sample estimate will be biased. Note that the forecast errors will include this in the error calculation. A similar argument holds for structural breaks in the growth rate of the time series: A break in the testing phase will likely lead to the exclusion of the predictor from the variable selection. A break in the testing phase will increase the measured forecast error. Since the PLS model is re-estimated at every step of the expanding window analysis, even this structural break will impact final forecasts for only a very limited number of observations. As the training window expands, the structural break will move from the testing to the training phase, and the PLS model will assign a weight of zero to the predictor time series with a structural break.

### 5.2. Robustness to time of forecast

This subsection limits the most recent AIS observations to the 15th day of a month to test the robustness of the forecast to a different time cutoff. While previous results rely on observed vessel positions and port calls up to the 30th day of the month, the cutoff date can be set to any day of the month. For example, one may aggregate ship positions from September 15 to October 15 and assign the observation to October for the training and testing phases of the algorithms. This allows publication of the PLS nowcasts for October in the second half of October. As Tables 24 and 26 show, the absolute forecast errors for the sample economies rise slightly. The forecast errors for imports rise from 2.2 pp (30-day cutoff, Table 24) to 2.8 pp (15-day cutoff) for the United States. The rise of forecast errors for the other highly developed countries is typically below 0.5 pp. The forecast errors for exports increase to a slightly higher extent. While relative RMSFE values also remain below one and signal that the PLS model continues to outperform benchmark models, there are weaknesses in the export forecasts. The benchmark produces lower forecast errors for nowcasts of EU exports, and both the standard deviation and benchmark RMSFEs are lower than forecast errors for French exports. These results do not carry statistical significance, and the PLS model still contributes unique predictive power to the forecasts of European exports according to the rejection of the null hypothesis $H_0^{PLS} : \lambda_1 = 0$ in Table 27. The benchmark model encompasses the PLS forecasts for French exports.

### 5.3. Robustness to price adjustment of trade flows

The second robustness check concerns the price adjustment of the dependent variable. The previously estimated unilateral trade flows are seasonal and price-adjusted from CPB (2022). In contrast, this subsection repeats the forecasting exercise for unilateral, nominal trade flows from the WTO (2022). The dependent variables are also seasonally adjusted using the X-13 ARIMA method. Tables 28 and 29 show the benchmark comparisons and encompassing tests for nominal imports. Absolute forecast errors tend to fall slightly when forecasting nominal imports. This is somewhat surprising since container shipping loads do not explicitly record the prices of the transported goods. It is nevertheless conceivable that a rising price level, in general, is correlated with the choice of transport mode. The PLS algorithm may detect such a correlation. Tables 30 and 31 show the respective export results. Overall, estimation errors tend to fall for nowcasts of nominal exports compared to real exports.

### 5.4. Robustness to benchmark model specifications

This study prominently features the ARIMA-ADL model as a benchmark for the performance of the PLS model and the AIS-derived predictor time series. Surprisingly, the benchmark model performs poorly for many countries, and the historical average growth rate is frequently a better predictor of future growth rates than the benchmark model. One explanation is that trade flows, particularly month-over-month growth time series with high variance, are difficult to predict. One would conclude that the sparse PLS model produces reasonable forecasts *despite* the fact that dependent time series are difficult to predict. A more disconcerting problem would be if the ARIMA-ADL model is incorrectly specified and the PLS results appear reasonable compared to a misspecified model. The following three robustness checks test the results' sensitivity to other benchmark model specifications.

Fig. 6 shows that the ARIMA terms lead to relatively poor predictions during COVID-related disruptions. As a result, this robustness check tests if the benchmark model improves when omitting the ARIMA model entirely and the autoregression terms from the ADL model. Hence, this robustness check only compares the PLS indicator against a model comprising other exogenous indicators. The regression formula for the benchmark model, therefore, is defined as in Eq. (10) without the autoregression terms:

$$y_{t+h} = \alpha_2 + \sum_{i=0}^{2}\sum_{j=0}^{n} \delta_{ij}\, MACRO_{t-i,j} + \sum_{i=0}^{2} \gamma_i\, SHIP_{t-i} + \sum_{i=0}^{2}\sum_{r=0}^{m} \phi_{ir}\, GLOBAL_{t-i,r} + \varepsilon_{t+h} \tag{22}$$

Table 38 shows the results for imports, while Table 39 displays the results for exports. The relative RMSFE values respond differently to the omission of autoregressive terms. For instance, the relative RMSFEs of the United

States imports and exports decrease. This points to the fact that the autoregressive terms, particularly the ARIMA model, did improve the benchmark model. In contrast, the respective relative RMSFEs for the United Kingdom do increase. In this case, structural breaks in the autoregressive terms may have decreased the forecasting power of the benchmark model.

Second, several leading indicators are published with a significant time lag. For instance, the maritime cargo loads calculated by Cerdeiro et al. (2020) and the RWI/ISL throughput index for ports are published with a time lag of approximately a month. Hence, the researcher would have to wait until at least early December to calculate the "nowcast" for the preceding October. The PLS model can instead calculate a nowcast for October at any time in the second half of October or early November. Several competing leading indicators are lagged by one period as outlined in Section 2 to reflect this time advantage. While this reflects reality, one can calculate a robustness check without this additional time. The robustness check, therefore, simulates how well the benchmark model would forecast if all leading indicators were available simultaneously as the PLS nowcasts. Tables 40 and 41 show the results for unilateral, real import time series, and Tables 42 and 43 show the respective results for unilateral, real export time series. The relative RMSFEs between PLS and benchmark models increase, which points to improved benchmarks. The PLS technique outperforms the benchmark model for export forecasts in $h = 0$ for 3 out of 11 selected countries, down from 6 out of 11 for the original specification.

In a final robustness check, the benchmark model omits the maritime cargo loads calculated by Cerdeiro et al. (2020) and disseminated by the IMF. Due to the focus on maritime cargo, the estimates are a natural benchmark for this study. Unfortunately, this data source started in June 2015, reducing the training period for the ARIMAX-ADL model. This robustness check omits the IMF data, increasing the training period for many countries. Tables 44 and 45 report the results for imports, whereas Tables 46 and 47 report the results for exports. In line with the previous robustness checks, the relative forecast errors of the PLS system against the benchmark increase. The PLS indicator can again only outperform the benchmark model for 3 out of 11 countries with statistical significance, down from 6 when including the IMF indicator. The drop is even higher for imports as the number of countries with statistical outperformance drops from 9 to 3 in the small selection.

### 5.5. Other robustness checks

Tables 36 and 37 show estimation errors when performing the leave-one-out out-of-sample exercise instead of the expanding window analysis. Despite the possibility that the LOO approach may prematurely incorporate future structural breaks, RMSFEs between the out-of-sample estimations are remarkably similar. Lastly, this study relies on the RMSFE loss function to measure the forecasting error. Since this study aims to develop tools to forecast extraordinary shocks, the RMSFE loss function is chosen due to its sensitivity to outliers. The forecast analyses are repeated using the mean absolute forecast error (MAFE) to assess the performance of the sparse PLS setup without emphasizing outliers. Results are presented in Online Appendix Tables 48 and 49. The respective third columns show that the relative MAFE against the ARIMAX-ADL benchmark ranges between 0.293 for world and 0.711 for Sub-Saharan imports. This compares to 0.651 for Brazilian imports and 0.808 for United States imports in the baseline estimation presented in Table 1. Overall, the relative MAFE lies below the relative RMSFE against the ARIMA-ADL benchmark. All relative forecast errors for the selected countries lie below 1 with a considerable margin. This indicates that the LASSO PLS setup performs better than the benchmark model independently of the loss function used.

## 6. Conclusion

This paper introduces methods to convert high-frequency, geocoded data from the container ship network into predictor time series, which is useful for economic forecasting. Since these methods can derive 880 or more independent time series, the paper combines a least absolute shrinkage and selection operator (LASSO) with a partial least squares model to reduce the dimensionality of the predictor space. The model is then used to forecast unilateral and bilateral trade flows for the current time period and one period ahead. Forecasts frequently outperform and encompass benchmark models with competing leading indicators for 76 countries and regions. Extensions show that these results hold for real and nominal trade flows, for developed and developing nations, and in times of economic crisis during the COVID-19 pandemic. Factor models, such as PLS, are well suited for an application to the shipping data and gain predictive power when combined with LASSO. Furthermore, overlaying the raw shipping data with a forecasting model has several advantages: All predictions made by the sparse PLS technique can be readily interpreted as monthly growth of national trade flows instead of leading indicators that measure the absolute cargo volume arriving or departing ports. The prediction model also exploits sources of correlation between a country and foreign ports, such as input–output linkages between countries. Extensions of this work may include applying the PLS system to other sources of transportation data, such as flight positions or truck tolls. Additionally, the currently available methods and data can be trained on other dependent time series beyond national headline trade figures, such as the economic activity of subnational regions or specific industries.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The author has previously been employed at the University of Kiel.

## Appendix A

See Tables 9–11.

**Table 9**
Variables for estimation of US-American imports in current month.

| Course/Orientation | Description |
|---|---|
| Eastern course | Western Panama Canal |
| Southern course | Western Mexican Coast |
| Northern course | Gulf of Mexico |
| Northern course | Central East Coast USA |
| Northern course | Northern East Coast USA |
| Western course | Strait of Gibraltar |
| Southern course | West Coast of Great Britain |
| Northern course | North Sea |
| Northern course | Java Sea |
| Southern course | Southern Chinese Coast |
| Arrival | Miami, United States |
| Arrival | New York, United States |
| Arrival | Freeport, Bahamas |
| Arrival | Cartagena, Colombia |
| Arrival | Hamburg, Germany |
| Arrival | Kiel, Germany |
| Arrival | Dunkirk, France |
| Arrival | Trieste, Italy |
| Arrival | Ain Sukhna, Egypt |
| Arrival | Chennai, India |
| Arrival | Tokyo, Japan |
| Arrival | Kobe, Japan |
| Departure | Jackson, United States |
| Departure | All ports, United States |
| Departure | Sepetiba, Brazil |
| Departure | Colon, Panama |
| Departure | Halifax, Canada |
| Departure | Bremen, Germany |
| Departure | Hamburg, Germany |
| Departure | La Spezia, Italy |
| Departure | Larvik, Norway |
| Departure | Chennai, India |
| Departure | Dong Guan, China |

The order of variables does no reflect the variable importance in the final estimation of the PLS.

**Table 10**
Overview of leading indicators used in benchmark model,(I/II)

| Country | Leading indicators |
|---|---|
| Argentina | NBS PMI MANUF, Argentina metr. t. - IMF, US IP, GSPCI - NY Fed<br>Argentina Business Survey (Argentina National Stats) |
| Australia | Australia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Australia PMI(Australian Industry Group), Australia New Orders(ACCI) |
| Austria | Austria COF, Austria ESI, US IP, GSPCI - NY Fed<br>Austria FP (DG ECFIN), Austria New Orders (Statistics Austria) |
| Belgium | Belg. COF, Belg. ESI, Belg. metr. t. - IMF, US IP, GSPCI - NY Fed, Northrange RWI/ISL<br>Belgium FP(OECD), Belgium PROD EXPECT(DG ECFIN) |
| Brazil | Brazil metr. t. - IMF, US IP, GSPCI - NY Fed<br>Brazil FP(OECD), Brazil INDUSTRIAL CONFIDENCE(OECD) |
| Bulgaria | Bulgaria COF, Bulgaria ESI, Bulgaria metr. t. - IMF, US IP, GSPCI - NY Fed<br>Bulgaria FP(OECD), Bulgaria PROD EXPECT(DG ECFIN) |
| Canada | Canada metr. t. - IMF, US IP, GSPCI - NY Fed<br>Canada PMI(Richard Ivey BS), Canada Business Surveys(CFIB) |
| China | China metr. t. - IMF, US IP, GSPCI - NY Fed<br>China PMI(NBS China),China New Orders (National Bureau of Statistics of China) |
| Croatia | Croatia COF, Croatia ESI, Croatia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Croatia Retail Trade Survey(DG ECFIN) |
| Cyprus | Cyprus COF, Cyprus ESI, Cyprus metr. t. - IMF, US IP, GSPCI - NY Fed<br>Cyprus PROD EXPECT(DG ECFIN), Cyprus INDUSTRIAL CONFIDENCE(DG ECFIN) |
| Czech Republic | Czech Republic COF, Czech Republic ESI, US IP, GSPCI - NY Fed<br>Czech Republic PROD EXPECT(DG ECFIN), Czech Republic INDUSTRIAL CONFIDENCE(DG ECFIN) |
| Denmark | Denmark COF, Denmark ESI, Denmark metr. t. - IMF, US IP, GSPCI - NY Fed<br>Denmark PMI(S&P Global), Denmark PROD EXPECT(DG ECFIN) |
| EU27 | EU27 COF, EU27 ESI, US IP, GSPCI - NY Fed<br>EU27 FP(DG ECFIN) |

(*continued on next page*)

**Table 10** (*continued*).

| Country | Leading indicators |
|---|---|
| EURO | EURO COF, EURO ESI, EURO metr. t. - IMF, US IP, GSPCI - NY Fed<br>EURO FP(DG ECFIN) |
| Estonia | Estonia COF, Estonia ESI, Estonia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Estonia PROD EXPECT(DG ECFIN), EURO INDUSTRIAL CONFIDENCE(DG ECFIN) |
| Finland | Finland COF, Finland ESI, Finland metr. t. - IMF, US IP, GSPCI - NY Fed<br>Finland FP(OECD), Finland PROD EXPECT(DG ECFIN) |
| France | France COF, France ESI, France metr. t. - IMF, US IP, GSPCI - NY Fed<br>France FP(OECD), France PROD EXPECT(DG ECFIN) |
| Germany | DE COF, DE ESI, DE metr. t. - IMF, US IP, GSPCI - NY Fed<br>Germany FP(OECD), Germany PROD EXPECT(DG ECFIN) |
| Greece | Greece COF, Greece ESI, Greece metr. t. - IMF, US IP, GSPCI - NY Fed<br>Greece FP(OECD), Greece PROD EXPECT(DG ECFIN) |
| Hong Kong, China | Hong Kong, China metr. t. - IMF, US IP, GSPCI - NY Fed |
| Hungary | Hungary COF, Hungary ESI, US IP, GSPCI - NY Fed<br>Hungary PMI (HALPIM),Hungary FP(OECD) |
| Iceland | Iceland metr. t. - IMF, US IP, GSPCI - NY Fed<br>Iceland Consumer Survey(Gallup) |
| India | India metr. t. - IMF, US IP, GSPCI - NY Fed<br>India PMI (Reuters),India Composite Leading indic.(OECD) |
| Indonesia | Indonesia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Indonesia Industrial Confidence(OECD) |
| Ireland | Ireland COF, Ireland ESI, Ireland metr. t. - IMF, US IP, GSPCI - NY Fed<br>Ireland FP(OECD), Ireland PROD EXPECT (DG ECFIN) |
| Italy | Italy COF, Italy ESI, Italy metr. t. - IMF, US IP, GSPCI - NY Fed<br>Italy FP(OECD), Italy PROD EXPECT (DG ECFIN) |
| Japan | Japan metr. t. - IMF, US IP, GSPCI - NY Fed<br>Japan Business Sentiment(Japan Finance Corporation for Small and Medium Enterprise) |
| Korea, Republic of | Korea, Republic of metr. t. - IMF, US IP, GSPCI - NY Fed<br>Korea, Republic of FP(OECD), Business Surveys (FKI-The Federation of Korean Industry) |
| Latvia | Latvia COF, Latvia ESI, Latvia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Latvia FP(OECD), Latvia PROD EXPECT (DG ECFIN) |
| Lithuania | Lithuania COF, Lithuania ESI, Lithuania metr. t. - IMF, US IP, GSPCI - NY Fed<br>Lithuania FP(OECD), Lithuania PROD EXPECT (DG ECFIN) |
| Luxembourg | Luxembourg COF, Luxembourg ESI, US IP, GSPCI - NY Fed |
| Malta | Malta COF, Malta ESI, Malta metr. t. - IMF, US IP, GSPCI - NY Fed<br>Malta Export Orders(DG ECFIN) |
| Mexico | Mexico metr. t. - IMF, US IP, GSPCI - NY Fed<br>Mexico PMI(INEGI), Mexico New Orders (INEGI) |

*Note:* metr. t. - IMF, refers to maritime export or import in metric tons as calculated by IMF. COF and ESI refer to the industrial confidence and economic sentiment indicators by the European Commission. US IP refers to the industrial product of the United States. GSCPI refers to the global supply chain pressure index by the New York Federal Reserve. Northrange RWI/ISL refers to the Northrange container throughput indicator by RWI/ISL. FP refers to future production survey and PMI refers to purchasing manager index.

**Table 11**
Overview of leading indicators used in benchmark model (II/II).

| Country | Leading indicators |
|---|---|
| Netherlands | NL COF, NL ESI, NL metr. t. - IMF, US IP, GSPCI - NY Fed, Northrange RWI/ISL<br>Netherlands FP(OECD), Netherlands PROD EXPECT(DG ECFIN) |
| New Zealand | New Zealand metr. t. - IMF, US IP, GSPCI - NY Fed<br>New Zealand PMI(S&P Global) |
| North Macedonia | North Macedonia COF, North Macedonia ESI, US IP, GSPCI - NY Fed<br>North Macedonia Industrial Production (State Stat. N. Maced) |
| Norway | Norway metr. t. - IMF, US IP, GSPCI - NY Fed<br>Norway PMI(NIMA/DNB Markets), Norway New Orders(NIMA/DNB Markets) |
| Poland | Poland COF, Poland ESI, Poland metr. t. - IMF, US IP, GSPCI - NY Fed<br>Poland FP(OECD), Poland PROD EXPECT(DG ECFIN) |
| Portugal | Portugal COF, Portugal ESI, Portugal metr. t. - IMF, US IP, GSPCI - NY Fed<br>Portugal FP(OECD), Portugal PROD EXPECT(DG ECFIN) |
| Romania | Romania COF, Romania ESI, US IP, GSPCI - NY Fed<br>Romania FP(OECD), Romania PROD EXPECT(DG ECFIN) |
| Russian Federation | Russian Federation metr. t. - IMF, US IP, GSPCI - NY Fed<br>Russian Federation FP(OECD), Russian Federation Business Confidence(Federal State Statistics Service, Russia) |
| Saudi_Arabia | Saudi_Arabia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Saudi_Arabia PMI (S&P) |

(*continued on next page*)

**Table 11** (*continued*).

| Country | Leading indicators |
|---|---|
| Singapore | Singapore metr. t. - IMF, US IP, GSPCI - NY Fed<br>Singapore PMI(S&P Global) |
| Slovak Republic | Slovak Republic COF, Slovak Republic ESI, US IP, GSPCI - NY Fed<br>Slovak Republic FP(OECD),Slovak Republic PROD EXPECT (DG ECFIN) |
| Slovenia | Slovenia COF, Slovenia ESI, Slovenia metr. t. - IMF, US IP, GSPCI - NY Fed<br>Slovenia PROD EXPECT (DG ECFIN) |
| South Africa | South Africa metr. t. - IMF, US IP, GSPCI - NY Fed<br>South Africa Busniess Expect (BER, South Africa) |
| Spain | Spain COF, Spain ESI, Spain metr. t. - IMF, US IP, GSPCI - NY Fed<br>Spain PMI(Reuters),Spain FP(OECD), Spain PROD EXPECT(DG ECFIN) |
| Sweden | Sweden COF, Sweden ESI, Sweden metr. t. - IMF, US IP, GSPCI - NY Fed<br>Sweden PMI(Swedbank and Silf),Sweden FP(OECD),Sweden PROD EXPECT(DG ECFIN) |
| Turkey | Turkey COF, Turkey ESI, Turkey metr. t. - IMF, US IP, GSPCI - NY Fed<br>Turkey FP(OECD) |
| United Kingdom | UK COF, UK ESI, UK metr. t. - IMF, US IP, GSPCI - NY Fed<br>United Kingdom PMI(Reuters), United Kingdom FP(OECD),United Kingdom Expect Output(CBI) |
| United States | United States metr. t. - IMF, US IP, GSPCI - NY Fed<br>United States PMI(ISM - Institute for Supply Managemen) ,United States New Orders |
| World | World metr. t. - IMF, US IP, GSPCI - NY Fed<br>World PMI(S&P Global) |
| Latin America | Latin America Industrial Production(Banco de la Republica CO) |
| Sub.S Africa | PMI( Central Bank of Nigeria) |
| South East Asia | Industrial Production(OIE Thailand) |

*Note:* metr. t. - IMF, refers to maritime export or import in metric tons as calculated by IMF. COF and ESI refer to the industrial confidence and economic sentiment indicators by the European Commission. US IP refers to the industrial product of the United States. GSCPI refers to the global supply chain pressure index by the New York Federal Reserve. Northrange RWI/ISL refers to the Northrange container throughput indicator by RWI/ISL.

## Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2023.11.007.

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